Isometric Standard

v2.0

The Isometric Standard lists the requirements that ensure delivered tonnes have measurable climate impact. Tonnes must be additional and calculated using the latest scientific techniques. These tonnes must also include a scientifically robust uncertainty quantification.

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1.0

Introduction

The Isometric Registry records Verified Credits that represent emission impacts which satisfy the latest Version of the Isometric Standard. This includes net Removals of atmospheric carbon dioxide equivalent (CO_2e) and Reductions of specified greenhouse gas emissions. The Isometric Standard, detailed in this document, lists the requirements that ensure Delivered Credits have measurable climate impact. In summary, Credits must represent a Removal or Reduction, be Additional and be calculated using the latest scientific techniques. This includes a scientifically robust Uncertainty quantification meeting at minimum with the requirements outlined in the Standard.

Isometric coordinates quantification and Verification necessary to Issue Credits on the Isometric Registry. All data, calculations and evidence associated with Credits are publicly available.

This Isometric Standard sets out the duties and obligations of stakeholders in relation to the Isometric Registry. Its purpose is also to provide for a consistent and predictable experience for all parties engaging with the Isometric Registry.

This Standard and the rules contained herein were designed and developed by Isometric HQ Limited.

1.1

Language guide

Throughout this Standard, and throughout Isometric Protocols, the words below are to be interpreted as follows:

- “Must” indicates a requirement.
- “Should” indicates a recommendation.
- “May” indicates a permissible course of action.
- “Will” indicates standard operating procedures implemented by Isometric.

1.2 Purpose and Principles

1.2.1 Purpose

The Isometric Crediting Program is guided by the Isometric Standard, and exists to:

- provide guidance, transparent infrastructure and to set a high scientific caliber that fosters high quality climate action in the form of durable net Removals of CO_2e and Reductions of greenhouse gas emissions; and
- Issue Verified Credits that can be used as an evidentiary statement of their ownership for Removal or Reduction claims and reporting purposes.

The Isometric Standard sets requirements that all Isometric Protocols must meet at a minimum.

1.2.2 Principles

The Isometric Crediting Program adheres to the following principles:

Collaboration

- Isometric engages with relevant governmental and Non-Governmental Organizations (NGOs) to facilitate equitable representation of stakeholders in a predominantly private industry and advance the creation of high quality industry standards.
- Isometric engages the scientific community and collaborates with experts in areas such as sensor development, physical inspection, auditing, earth-system modeling and data analysis to Verify Deliveries.
- Where appropriate and practicable, Isometric Systems may be integrated with similar tracking systems to help ensure full and accurate information for Buyers of Removals and Reductions and minimize the potential for Double Counting.

Incentive Alignment

- Isometric does not sell or broker the sale of credits.
- Isometric charges Buyers – not Project Proponents – a single flat fee per Offtake or purchase which is based on effort required for monitoring, reporting, and verification of a specific pathway/process. This price structure is independent from the number of Credits issued and thus reduces incentives for overcrediting.
- Isometric collects fees from Buyers on a payment schedule decoupled from the credit delivery schedule. This mechanism eliminates incentives for early crediting.
- Isometric does not receive payments from Project Proponents to Verify the delivery of tonnes and adherence to Certified Protocols.
- Isometric undertakes the cost of Protocol development and third-party Verification to ensure neither Buyers nor Project Proponents can have undue influence over this process.

- Isometric independently and impartially chooses third-party Validation and Verification Bodies (VVBs) for Projects.

Scientific Rigor

- Isometric uses the latest science to quantify net Removal or Reduction, Durability, Uncertainty and Additionality across a variety of heterogeneous Removal and Reduction Pathways.
- Isometric will review and update Protocols without a penalty on previously Issued Credits in order to incentivize continued investigation of underlying scientific questions – through which Removal or Reduction estimates can be further constrained.
- Isometric quantifies all Carbon Fluxes and Greenhouse Gasses related to the Removal or Reduction activity taken by the Project Proponent in a Conservative manner that incorporates uncertainty – including second order effects on leakage.

Transparency

- Isometric provides Buyers and Beneficiaries with access to full traceability of Carbon Fluxes involved in the quantification of Removals or Reductions, alongside the evidence gathered for the Validation and Verification Process. For Reductions, gross reduction is reported in CO_2e and broken down into avoided fossil CO_2 and avoided non-CO_2 greenhouse gasses.
- Protocols and quantification Models are transparently versioned, allowing for public tracking of changes, and public consultation after major updates.
- Each Credit Issued by Isometric is publicly accessible – allowing for inspection of the information used to quantify Removals or Reductions.

1.3
Scope

1.3.1
Project Type

Isometric works with Projects conforming to a Protocol which has undergone or will undergo the Isometric Certification Process. Projects are not eligible if:

- They do not meet the conditions to justify the minimum Durability set out in the applicable Protocol or Module, or
- Project activities lead to Enhanced Hydrocarbon Recovery (EHR).

1.3.2
Credit Type

Isometric Issues two Types of Credits:

- Carbon Dioxide Removal credits, for the net removal or one metric tonne of CO_2e, and
- Emission Reduction credits, for the net reduction of one metric tonne of CO_2e of specified greenhouse gasses generated by an existing emitter.

Projects can issue multiple Types of credit.

1.3.3
Geographic Scope

Isometric works with Projects worldwide. Isometric works in English. Additional languages may be considered given

translation by a trusted source.

In some instances, Modules within Protocols may be tailored for Projects which have geographical specificities. In this circumstance, geography-specific alterations to utilized Modules will be made clear in order to maintain transparency amongst Project Proponents.

1.3.4
Greenhouse Gas (GHG) Eligibility

Isometric issues Removal and Reduction credits. The following greenhouse gasses (GHGs) are relevant:

- For Carbon Dioxide Removal credits, the removal of atmospheric CO_2,
- For Emission Reduction credits, reductions of specified greenhouse gasses generated by an existing greenhouse gas emitting source or activity, as defined in the relevant Protocol.

In evaluating the comprehensive greenhouse gas (GHG) emissions of a Project, all GHGs are considered. Isometric uses the United States Environmental Protection Agency definition of GHGs[1], which includes: carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and fluorinated gases such as hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF_6), and nitrogen trifluoride (NF_3).

1.4
Versioning

This Isometric Standard was first published 4th October 2023.

This document will be updated on an ongoing basis to reflect changes in the operation, governance and/or rules that apply to either Projects on, or activity described by, the Registry.

Any proposed material changes to the Standard will be put forward for review by the independent Science Network. Following any amendments resulting from that process, the draft changes will be put out for public comment. After having incorporated any further amendments, the final changes will be published in an updated version of the Standard, alongside a clear explanation of the stakeholder input received and how this informed the final changes that were made.

Version numbers follow the MAJOR.MINOR.PATCH format, where three non-negative integers denote the version of the Standard.

Incrementing each number signifies a different type and magnitude of change:

- **Major** – Material changes significantly altering the scope, applicability, or foundational framework of the Standard.
- **Minor** – Material changes that may have significant impact on Project eligibility, Verification practices, or Credit issuance.
- **Patch** – Improvements without introducing material changes, such as fixing typographic errors, clarifying wording, or adding examples.

A full changelog, along with all previously published versions and their dates of publication, is publicly available on Isometric's website.

1.4.1

Standard version adoption

New versions of the Standard follow the same adoption requirements as the ones outlined for the Protocol version adoption in Section 2.4.5.

1.5

Citation

The appropriate citation for the document is the Isometric Standard (2026), version 2.0.

2.0

Protocol Requirements

This section outlines the requirements for all Isometric Certified Protocols including Isometric Certified Modules.

2.1

Protocol Certification Process

All Projects must be Validated against, and Removals and Reductions Verified against, an Isometric Certified Protocol.

An Isometric Certified Protocol is a Protocol developed by Isometric that has undergone Public Consultation, and is listed as Certified on the Isometric Registry. Protocols may incorporate Modules, using the Isometric Modular Framework.

Isometric will not list multiple Protocols or Modules as Certified where they pertain to the same or essentially the same underlying carbon removal activities.

2.2

Modular Framework

A Protocol can incorporate an existing Isometric Certified Module. Modules are developed by Isometric and follow the same Certification process.

Each Protocol minor version specifies and locks in the exact minor versions of all Modules which it incorporates. Once certified, this combination never changes, so the Protocol minor version number alone specifies which Module versions apply.

All Protocols will incorporate the same Modules for GHG accounting, in order to create a consistent accounting

framework across all Projects.

Isometric will review individual Modules annually and be responsible for version changes in accordance with our Updates to Protocols policy.

2.3 Consultation Requirements

The external Consultation process will be as follows:

1. Expert review through the Isometric Science Network
2. Public Consultation

Expert review will be conducted through the Isometric Science Network. Expert review is the process in which the internal Isometric Science Team invites individuals with deep subject matter expertise to submit feedback on a Protocol or Module. Isometric generously compensates experts for their detailed review with a flat hourly fee paid upon feedback submission.

Public Consultation will be announced on the Isometric Registry and to the Isometric Science Network. Protocols and Modules are open for public commenting for a minimum of 30 days. All relevant stakeholder comments are considered and responded to by the internal Isometric Science Team, and the results of the Consultation will be published on the Registry. All anonymized comments and reviews will be made available upon request.

Typically, at least 5-10 scientific experts are engaged for a given Module or Protocol via the combination of the above processes.

When a Protocol is updated and the changes to it are limited to introducing, upgrading, or removing Modules that have already undergone Consultation, the Module Consultations fulfill the Protocol Consultation requirement. A Module Consultation must address the impact of the Module on any Protocol that will incorporate it.

2.4
Updates to Protocols

2.4.1
Protocol update schedule

The internal Isometric Science Team will review a Protocol or Module if Isometric deems it necessary due to material changes in realms such as scientific knowledge, technology and/or regulatory frameworks. This review will be completed within a period of 6 months from the date an issue is raised. If the review results in substantial changes to a Protocol or Module such that a new version is required, these changes will adhere to the Protocol Certification process.

In addition to the triggers for review outlined above, each Protocol will be reviewed at the sooner of the following criteria:

- after 2 years have passed since the original Certification; or
- whenever the number of Credits Issued under a Protocol passes the following milestones: 100,000 Credits Issued; 500,000 Credits Issued; 1,000,000 Credits Issued; 5,000,000 Credits Issued.

Major and minor changes will be documented on the Isometric Registry and published alongside the results of a public commenting period of a minimum duration of 30 days.

All versions of Protocols and Modules will be documented, archived and made publicly available. All revisions and updates will be systematically versioned according to the Protocol version number format.

Isometric reserves the right to suspend and/or withdraw a Protocol if deemed necessary due to credible, evidence-based concerns that carbon removal activities are being overestimated or that Additionality may not be ensured. In such exceptional cases, Isometric may take immediate action to suspend and/or withdraw a Protocol. The Protocol may then be re-introduced following the usual period of public consultation in the case of any material change.

2.4.2

Impact of Protocol updates on Projects and Credits

Credits issued against a previous Protocol version are not affected by Protocol updates, unless otherwise specified with justification in the updated Protocol. Projects must update to newer Protocol versions according to Protocol adoption requirements, with continuity of operations protected by Protocol backward compatibility mechanisms.

2.4.3

Protocol update materiality

For the purposes of Protocol versioning, material changes are defined as those which affect how Validations and Verifications are conducted, or their outcomes. If a change requires Validation and Verification Bodies to modify their approach, methods, or decisions, it is considered material.

2.4.4

Protocol version number

Version numbers follow the `MAJOR.MINOR.PATCH` format, where three non-negative integers denote the version of the Protocol or Module. Incrementing each number signifies a different type and magnitude of change:

- **Major** – A significant departure from a previous version, introducing material changes to validation and verification rules. Typically used to remove support for old practices.
- **Minor** – An evolution from a previous version, introducing material changes to validation and verification rules. Typically used to introduce support for new practices, adapt to scientific or legal developments, or systematize existing rules.
- **Patch** – An improvement without introducing material changes. For example – fixing typographic errors, clarifying wording, or adding examples.

A full changelog for each Protocol, along with all previously published versions and their dates of publication, is publicly available on Isometric's website.

2.4.5

Protocol adoption requirements

New Projects must validate against the most recent certified major and minor version of the relevant Protocol. Projects may qualify for an earlier version by commencing Validation or initiating Project registration on the Registry prior to the new version's certification. Isometric may also, at its discretion, permit a Project Proponent to validate a Project against an earlier Protocol version in other scenarios; for

example if the most recent version has not been approved by ICVCM or CORSIA.
The adoption requirements upon release of a new Protocol version are as follows:

- **Major** – Projects are not required to adopt a new major version until they seek renewal of their Crediting Period.
- **Minor** – Same as Major: Projects are not required to adopt a new minor version until they seek renewal of their Crediting Period.
- **Patch** – Take effect automatically upon release. Projects operate under the latest patch version of their current Protocol minor version.

Isometric reserves the right to issue a mandatory adoption schedule in exceptional circumstances in order to resolve a substantive vulnerability, or due to material changes in realms such as scientific knowledge, technology and/or regulatory frameworks.

2.4.6 Protocol backward compatibility mechanisms

When new Protocol versions are Certified, Projects can continue operating uninterrupted through the use of the following backward compatibility mechanisms:

- **No forced major and minor version upgrades** – Projects are not required to adopt a new Protocol major or minor version until they seek renewal of their Crediting Period, or an exceptional mandatory adoption schedule was issued.
- **Legacy exemptions** – Project Proponents may voluntarily adopt a new version and request exemptions against new requirements introduced after the Protocol

minor version used for initial Validation in the current Crediting Period. Where a Project Proponent seeks such an exemption against specific requirements, the Project Proponent must demonstrate that said requirements create non-trivial operational difficulty or cost. Where granted, such exemptions remain valid until the end of the current Crediting Period.

- **Retained validation status** – Projects do not require revalidation when adopting a new Protocol minor version.

2.5 Protocol Contents

2.5.1 Applicability

All Protocols must include an Applicability section.

The Applicability section must, at a minimum, outline the activities to which the Protocol applies and whether Removal credits, Reduction credits, or both are generated from applicable Projects.

The Applicability section may also set out conditions that a project must meet in order to be eligible for crediting under the Protocol, including specific baseline requirements.

2.5.2 Boundaries

Protocols require Cradle-to-Grave GHG accounting of all emissions associated with a Project's Removal and/or Reduction process.

The GHG emissions that result from the Project's activities within the defined boundary, combined with any Leakages must together encompass the entire impact of a Project on GHG emissions. This must be presented as part of a GHG Statement and corresponding GHG Statement Report.

All Projects must have a defined temporal and geographic project boundary.

The system boundary for GHG accounting sets out the GHG Sources, Sinks, and Reservoirs associated with the project boundary and to be considered in the GHG Statement. The system boundary is specified by the relevant Protocol and must include at a minimum all GHG SSRs from:

- the construction or manufacturing of each physical site and associated equipment;
- the operation of each process; and
- the closure and disposal of each site and associated equipment.

2.5.3 Baselines

All Projects must be assessed against a Baseline scenario of their activities not having taken place.

Establishment of the Baseline scenario will differ between Project types and will be specified in the relevant Protocol.

A project's Counterfactual is:

- For a Removal, a quantification of the CO_2 removal and storage that would have taken place without the project's activities.
- For a Reduction, a quantification of the greenhouse gas reduction specified by the relevant Protocol that would have taken place without the project's activities.

This must be calculated using conservative assumptions, as outlined in relevant Protocols, and must be assessed for every Removal or Reduction.

Project Baselines must account for any existing government policies or legal requirements that lower GHG emissions.

Projects will only be Credited for Removals or Reductions above and beyond what was likely to have occurred in the Baseline scenario.

Projects must reassess and validate the Baseline scenario whenever they request a Crediting Period renewal.

2.5.4
Default Emission Factors, Proxies and Models

Protocols can use Default Emission Factors, Standards, Proxies and Models to quantify Removals or Reductions, demonstrate Additionality and/or establish Baselines. In metered systems, the preference is for direct measurement of required variables.

Proxies and Models will only be permitted for use under circumstances which are justifiable, such as for reasons of practicality, reliability, and for gap filling in variable un-metered systems.

This justification must be outlined in the Protocol. Proxy measurements are secondarily preferred, and models may

be used when neither Proxy or direct measurements are viable.

When used, Models and Proxy measurements must apply Conservative Uncertainty factors and make Conservative assumptions.

Assumptions and estimations must be disclosed and justified.

When Models and Proxy measurements are used, Uncertainty must be assessed and clearly outlined in the Protocol.

Further requirements and guidance for the use of Default Emission Factors, Standards, Proxies and Models in Protocols are outlined below.

2.5.4.1 Default Emission Factors

The Protocol must appropriately cite and describe the source of any third-party Default Emission Factor, Standard, Model or Proxy. Third-party Default Emission Factors and Standards must be either internationally recognized or derived from a Reputable Source.

Where a newly established Default Emission Factor is used in a Protocol, the reasoning behind this choice and documentation of Default Emission Factor calculations must be clearly outlined and will be subject to review in accordance with the Consultation Requirements.

Protocols must identify Default Emission Factors that are likely subject to future changes. Default Emission Factors can be subject to updates and revisions given the Isometric Updates to Protocols policy.

2.5.4.2 Proxies

Proxies must be shown to be well-correlated with the variable of interest, preferably through established calibration studies from Reputable Sources.

The Proxy data, empirical fits, correlation data, and sources must be provided to Isometric and verified at the project's next verification.

2.5.4.3 Models

Models must be from a Reputable Source and/or shown to be reliable via peer-review, testing or correlation with empirical data.

The source of models, any modifications, input parameters, data used and validation results must be clearly outlined to Isometric and verified at the project's next verification.

2.5.5 Common Calculation Factors

All calculations must use consistent, standardized factors, including the following:

Global Warming Potential (GWP): Unless otherwise stated in a Protocol or Module, calculations must use the 100-yr GWP for the GHG of interest, based on the most recent volume of the IPCC Assessment Report, currently the Sixth Assessment Report[2].

2.5.6 Uncertainty Accounting

Uncertainty is due to factors such as inability to precisely measure certain physical phenomena and/or assumptions in Models or GHG accounting. Protocols must set up frameworks that allow for the reduction of uncertainties with time as measurements, models, and scientific understanding improves.

2.5.7 Durability

Emission impacts can have a Risk of Reversal and a finite durability associated with them. Alternatively, some Reductions are permanent with no risk of reversal, for example when a high GWP gas is permanently converted to a low GWP gas.

Protocols and Modules will specify if an emission impact has a risk of reversal and finite durability with one or more Durability Threshold(s), or in the case of some Reductions if it is permanent. Where a Protocol or Module does not explicitly specify a Durability Threshold or that the emission impact is permanent, the default Durability Threshold for the Protocol or Module is 1,000 years.

Projects following a Protocol or Module crediting non-permanent emission impacts must select from the Durability Threshold(s) defined in the Protocol or Module to be the Project Durability Threshold, by specifying this in the Project's PDD.

Projects crediting non-permanent emission impacts must demonstrate a Durability in excess of the designated Project Durability Threshold.

Removals or Reductions will only be Credited for their impact in excess of the specified Project Durability Threshold.

Where Projects generating Removals or Reductions cause additional storage or reduction below the Project Durability Threshold for the Project, this storage or reduction cannot count positively towards the end number of Credits, but can be used to offset losses in Counterfactual storage or reduction below the Project Durability Threshold, as outlined in the relevant Certified Protocol.

2.5.7.1 Justification of Durability

Protocols and Modules crediting non-permanent emission impacts will set requirements which Projects must meet in order to justify the claimed Project Durability Threshold.

These requirements may include:

- Providing information and justification in the Project's PDD and during validation;
- Presenting scientifically falsifiable hypotheses in the Project's PDD;
- Applying an uncertainty discount in the Conservative estimate of Removal or Reduction to account for storage uncertainty;
- Conducting ongoing Monitoring, as described in Storage Monitoring.

2.5.7.2 Storage Monitoring

A full risk assessment must be undertaken to identify all possible mechanisms that will lead to Reversals of crediting

non-permanent emission impacts and subsequent decreases in Durability.

There must be a monitoring plan in place to quantify the amount of potential Reversal that may occur via each identified Reversal mechanism.

The duration of storage monitoring required is process and location specific and requirements will be specified in the relevant Protocol.

Monitoring requirements must include:

- adherence to the monitoring program of the Protocol that the individual Project is following;
- the frequency of measurement and reporting, as specified in the relevant Protocol;
- consideration of Baselines and incorporating provisions for reevaluation at the end of a Project's Crediting Period or at set timescales as defined within the Protocol;
- the methodology for detecting all potential Reversal mechanisms;
- provisions for reporting Reversals to the VVB and Isometric, as adequate deductions to net Removals or Reductions may be required;
- identification of (and actionable plan for remediation of) emissions of CO_2e during a Project's operational and post-cessation lifespan;
- monitoring reports that are made publicly available to the Registry; and
- reassessment of reversal risk using the risk of reversal questionnaire at the renewal of each crediting period, or sooner if:
    - the relevant Protocol requires a more frequent assessment,

- monitoring identifies a reversal-related risk,
- the Project storage technology materially changes,
- an actual reversal event takes place.

2.5.8 Risk of Reversal

Protocols must categorize the Risk of Reversal for determining the appropriate Buffer Pool Size.

For Projects whose Protocol or Module for Crediting does not have its own risk assessment, Isometric has provided a risk questionnaire (see Appendix C) that can serve as guidance on how Risk of Reversal is determined.

The answers to the Protocol or Module specific risk assessment, or the default risk questionnaire, are used to support the Risk of Reversal and Buffer Pool Size.

The Risk of Reversal and the corresponding Buffer Pool Size are meant to compensate for the risk of Reversals that may be observed as a result of monitoring. The Buffer Pool is not meant to compensate for reversals of greenhouse gasses that are stored in an open system (e.g., the ocean) and direct observation of reversals would not be possible. In open systems, storage risks must be assessed as part of the Uncertainty assessment and accounted for in the Conservative estimate of Removal or Reduction. As such, Protocols that store carbon in open systems typically fall into the “No observable risk” Risk of Reversal category, and correspondingly are not required to make buffer pool contributions, or to complete a reversal risk questionnaire. This also applies to Reductions with a permanent emission impact. See Buffer Pools for further information.

3.0

Project Requirements

Removals and Reductions are conducted and Verified as part of specific Projects. This section describes the requirements for Projects to be compliant with the Isometric Standard.

3.1

Ownership

Projects must demonstrate that they have legal ownership over the rights to all Removals and Reductions that will be claimed from the Project, including at minimum making a representation to Isometric to this effect.

When there are multiple parties involved in the Removal and/or Reduction process, and to avoid Double Counting, a single Project Proponent must be specified as the sole owner of the Removals and/or Reductions.

Ownership must be defined in contracts between the Project Proponent and other Project participants, which may include, for example, suppliers of Feedstocks, transportation and logistics companies, and/or storage site owners and operators.

Contracts must:

- relinquish ownership of all Removals and/or Reductions to the Project Proponent;

- clearly indicate that other companies or operators participating in the Project cannot claim Removal

and/or Reduction Credits as a result of participation in the Project;

- provide mechanisms for the Project Proponent to obtain required information to fully calculate the Removals and Reductions from aspects of the Project relevant to the company; and

- stipulate that unless these third parties are the end owners of the generated Credits that they will not advertise that they are producing a “low emission product or practice,” in connection with or benefitting from the Removals or Reductions carried out by the Project.

These requirements do not preclude the project activity from involving multiple underlying entities (e.g., subsidiaries, special purpose entities or site owners).

3.2 Documentation

For a Project to be evaluated for the Isometric Registry, the Project Proponent must document Project characteristics in a Project Design Document (PDD). The document will form the basis for Project Validation and evaluation in accordance with the relevant Certified Protocol. The PDD must be consistent with ISO 14064-2:2019 [3], and must include:

- Project title, purpose(s) and objective(s);

- type of Project, including descriptions of the technologies, products, services and infrastructure to be utilised and how the Project will achieve Removal

and/or Reduction based on the calculation requirements laid out in the relevant Protocol;

- Project location, including organizational, geographic and physical location information, allowing for the unique identification and delineation of the specific extent of the Project;

- conditions prior to Project initiation to support identification of Counterfactual removals or reductions;

- anticipated aggregated net Removals and aggregated Reductions likely to occur from the Project, in tonnes of CO_2e;

- anticipated Uncertainty determination approach;
- identification of risks that could substantially affect the Project's Removals and Reductions and, if applicable, any measures to manage those risks;

- roles and responsibilities, including contact information of the Project Proponent and other Project participants;

- a summary environmental impact assessment;
- description of stakeholder, community, or other interested party Consultations and mechanisms for ongoing communication, and outcomes of any such discussions;

- a chronological plan or actual dates and justification for the following:
    - Project Start Date;

    - Baseline time period;
    - Project Crediting Period; and

- a monitoring plan detailing the elements to be monitored, frequency of monitoring and reporting, including relevant Project activities in each step of the Project cycle, measurement equipment, methods and procedures, details on accuracy and calibration, the clear assignment of responsibilities of parties involved in monitoring and reporting to Isometric, and appropriate measures for quality assurance and quality control; and
- If the project is crediting an emission impact that is non-permanent, Project Risk of Reversal.
    - Note that project Risk of Reversal must be greater than or equal to the Risk of Reversal outlined in the relevant Protocol. The Risk of Reversal can be increased to account for site-specific considerations that may make a Reversal more likely (e.g., increased risk of a natural disaster). Any increase in Risk of Reversal must be documented, along with its justification, in the PDD;
- an assessment of the compatibility of the Project's Removal and/or Reduction activities with transition to net zero, by reference to the net zero objectives of the host country.

- a description of the data collection and storage approach taken

- the Project's system boundary and an outline of all GHGs considered across all sources, sinks and reservoirs (SSRs). Robust justification and appropriate evidence must be provided for any GHG SSRs that have been excluded from the project boundary.

Isometric will not publicly disclose sensitive business information inherent to data included in the PDD or otherwise provided by the Project Proponent. Information will be labeled as sensitive by the Project Proponent, and Isometric will communicate to the Project Proponent whether any information deemed sensitive is necessary to be shared in order to accurately reflect Removal and/or Reduction GHG calculations. Sensitive business information may include, but is not limited to:

- locations,
- names,
- proprietary processes,
- non-public acquisitions/partnership plans, etc.

This cannot prevent the disclosure of baselines and other underlying assumptions critical for robust quantification of Removals or Reductions.

3.3 Eligibility

Projects are only eligible to claim Credits for activities that are exclusively registered with the Isometric Registry. Projects are only eligible to receive Credits on the Isometric Registry using an Isometric Certified Protocol.

Prior to Project registration on the Isometric Registry, projects must undergo a 30 day public comment period. Isometric will review all comments received during that period, and share these with the Project Proponent and the Project's appointed VVB. In response to comments received, Isometric or the appointed VVB may issue clarification requests or corrective action requests to the Project Proponent during the validation process. Any such corrective

action requests must be addressed by the Project Proponent during the validation process. Isometric will publish a summary of comments received once Project validation is complete.

At time of Project application and Validation, Projects must use the latest available version of a Certified Protocol, unless a grace period has been explicitly specified by Isometric, whereby a former version of a Protocol may continue to be used for a defined time period. Where Validation occurs after Project application, the Protocol version required is that which was latest at the time of application.

Projects must conform to all relevant laws and regulations in the jurisdiction in which they operate.

Projects must be Additional, according to the guidance described in the Isometric Standard Additionality approach.

Projects must use appropriate Emission Factors as set out in relevant Protocols. Protocols and modules set guidelines for the types of emissions factors and sources of emission factors that are acceptable.

The Project Proponent is responsible for notifying Isometric of any changes to operations that could change the eligibility of their process. Following notification, the project's eligibility must be assessed at the project's next verification.

3.4
Project Crediting

Projects are eligible to begin submitting GHG Statements to Isometric following Project Validation. Once Removals or Reductions are Verified, they may be used to issue Credits. A

single project can issue Removal credits, Reduction credits or both. See Section 6 for more information on Crediting.

The initial Crediting Period Start Date for a Project designates the beginning of operational removal or reduction activity for the Project. The requested Crediting Period Start Date must be no earlier than 5 years before the initial submission of the Project to the Registry. GHG Statements may be submitted for activity which took place on or following the Crediting Period Start Date.

Projects are eligible to Issue Credits for the duration of the Crediting Period specified in the Project's PDD. If a Project Proponent wishes to renew the Crediting Period of a Project, an updated PDD must be provided and the Project must be re-Validated. There is no limit on the number of times a project can be renewed providing it is still in compliance with the relevant Certified Protocol. In general, the maximum Crediting Period is 10 years, unless otherwise specified by the relevant Certified Protocol. The only Certified Protocols which currently allow for a longer period are the Biogenic Carbon Capture and Storage Protocol and Direct Air Capture Protocol, which each have a 15 year Crediting Period, and Biosphere Protocols in which the Crediting Period is defined per project.

Unless otherwise specified in the applicable Protocol or Module, if a validated Project has not undergone a verification for more than 12 months since the previous verification, an updated PDD must be provided and the Project must be re-Validated.

Isometric does not Issue Ex-ante Credits.

3.5
Additionality

The Project Proponent must be able to demonstrate four pillars of Additionality (Financial, Common Practice, Environmental and Regulatory), in order to show that the claimed environmental impact would not have otherwise occurred in the Baseline Scenario.

A Project can be considered Additional when all of the following criteria are met.

- Financial:
  - The Project can be considered to demonstrate Financial Additionality if Removals and/or Reductions are the main purpose and only source of revenue of the Project.
  - Otherwise, the Project must demonstrate that economic barriers would prevent Project implementation in the absence of Carbon Finance, as outlined in Financial Additionality Considerations below.

- Common Practice:
  - The Project must demonstrate that activities similar to the activities of the proposed Project are not common practice, as outlined in Common Practice Analysis.

- Environmental:
  - For Removals, the Project can be considered to demonstrate Environmental Additionality if the climate impact of the Project is net negative after subtracting the Counterfactual CO_2 removal and all

project GHG emissions, including leakage, from Project CO_2 removals, in accordance with the assessment framework defined in the relevant Protocol.

    - For Reductions, the Project can be considered to demonstrate Environmental Additionality if the climate impact of the Project is a net emission reduction after subtracting the Counterfactual GHG reduction and all project GHG emissions, including leakage, from Project GHG reductions, in accordance with the assessment framework defined in the relevant Protocol.

- Regulatory:
    - The Project can be considered to demonstrate Regulatory Additionality if the Project is not required by any regulatory, policy or other legal requirement.
    - Otherwise, the Project must be able to demonstrate that it exceeds the minimum regulatory requirements, as outlined in the Regulatory Additionality Considerations section below.

3.5.1 Common Practice Analysis

As per the requirements in Additionality, Projects must demonstrate that activities similar to the activities of the proposed Project are not Common Practice, by following the framework outlined here.

Projects must successfully demonstrate Common Practice Additionality via one of the following approaches. The approach chosen must be described in the Project PDD:

1. Projects in pathways which are below Technology readiness level (TRL) 8 or 9, as shown in the table below,

may be considered as demonstrating Common Practice Additionality without further analysis, due to the level of maturity of the relevant technology in its specific application to carbon removal and/or emissions reductions.

- Project TRL must be determined in accordance with the table below; alternatively, a Project may submit its own assessment of TRL, provided that the assessed TRL is not lower than the corresponding provider TRL values set out in the table below. Where a Pathway TRL range spans values both above and below the threshold level of TRL 8, the Project must submit its own TRL assessment.

2. Or, Projects may complete a full Common Practice Analysis, as described below.

PATHWAY	TECHNOLOGY READINESS LEVEL (TRL)
Direct Air Capture (DAC)	6
Enhanced Rock Weathering (ERW)	3-4
Mineralization	6-7
BiCRS / Bio-CCS	5-6
Biochar*	6-9
Oceans and Rivers**	1-7
Reforestation	8-9
Landill Methane Flaring and Utilization***	9

Technology Readiness Levels (TRLs) of Pathways as defined by IPCC [4], and otherwise as estimated by Isometric. TRL of Biochar Projects may vary based on the maturity of the specific capture technology employed (e.g. biomass-boiler retrofits may indicate a higher TRL). TRL of Oceans & Rivers

taken as range estimated by IPCC[4] (1-2) and RMI[5] (4-7) to reflect the fact that Oceans & Rivers pathway has a wide range of TRLs depending on the individual technology in question.TRL of Landill Methane Flaring and Utilization taken as estimated by a report commissioned by the UK's Climate Change Committee (CCC)[6]

Projects following Option (2) must provide an analysis justifying that activities similar to the activities of the proposed Project are not Common Practice, by following the below requirements.

Projects must describe the applicable geographic area under consideration for the analysis. The default area is the country of Project operation, however the Project may propose a more specific region with justification, as outlined below.

If a Project proposes a more specific applicable geographic area, the Project must justify the essential distinctions between the applicable geographic area and the rest of the country of Project operation that give rise to different implementation conditions pertinent to the Project activity. Relevant factors may include:

- Differences in laws, regulations or policies;
- Geological or climatic differences;
- Economic or socioeconomic conditions;
- Market conditions or access to resources;
- Availability and accessibility of infrastructure;
- Labor force and expertise.

Project Proponents must select one of the following two methods for undertaking a Common Practice analysis, within the applicable geographic area:

- (i) For Projects deploying distributed technologies, such as enhanced rock weathering, or reforestation, proceed to “Method for assessing Common Practice for distributed technologies”.
- (ii) For Projects operating discrete facilities, or plants, proceed to “Method for assessing Common Practice for discrete facilities”.

3.5.1.1

(i) Method for assessing Common Practice for distributed technologies

- The Project must define market penetration, or Common Practice, in terms of recent uptake or existing stock/diffusion of technologies, services or practices in relation to a realistic market size.
- Identify the relevant population of subjects, such as a similar class of adopters or landowners, within the selected applicable geographic area.
- Where similar activities to the proposed Project activity are identified, compare the Project activity to these other activities. Identify and explain any essential distinctions between the proposed Project and similar activities. Essential distinctions may be related to, for example:
    - Differences in laws, regulations or policies;
    - Geological or climatic differences;
    - Economic or socioeconomic conditions;
    - Market conditions or access to resources;
    - Availability and accessibility of infrastructure;
    - Labor force and expertise;
    - Significant changes in circumstances, which affect the implementation of the Project activity, in comparison to similar activities.

- Calculate the cumulative market penetration rate (as a percentage) of the Project activity within the population who have not received carbon finance revenue (e.g., where activities are neither part of a registered Isometric Project, nor registered under other GHG programs), in the sample of adopters.
- If the market penetration rate is below 20%, the activity is considered to demonstrate Common Practice additionality.

3.5.1.2

(ii) Method for assessing Common Practice for discrete facilities

- Assess the extent to which similar activities to the proposed Project activity have been implemented previously or are currently underway in the applicable geographic area. Similar activities are those with comparable technologies or practices, scales, and regulatory environments.
- Exclude activities which are under validation or registered under the Isometric Standard; activities under other GHG programs may optionally be excluded. Record N_{all} as the total remaining number of similar projects or activities.
- Where similar activities to the proposed Project activity are identified, compare the Project activity to these other activities. Identify and explain any essential distinctions between the proposed Project and similar activities. Essential distinctions may be related to, for example:
    - Differences in laws, regulations or policies;
    - Geological or climatic differences;
    - Economic or socioeconomic conditions;
    - Market conditions or access to resources;
    - Availability and accessibility of infrastructure;

    - Labor force and expertise;
    - Significant changes in circumstances, which affect the implementation of the project activity, in comparison to similar activities.
- Identify the following parameters, and assess Common Practice additionality, as follows:
    - N_{all}, defined as the total number of similar projects in the applicable geographic area.
    - N_{diff}, defined as those projects which have essential distinctions from the proposed Project.
    - Calculate the "Common Practice factor" $F = 1-(N_{diff}/N_{all})$, which represents the share of similar projects without essential distinction in the applicable geographic area.
    - The proposed Project activity is considered to demonstrate Common Practice additionality where:
        - N_{all} is less than or equal to 3;
        - Or, F is below or equal to 20%.

3.5.2 Financial Additionality Considerations

As per the above requirements in Additionality, the requirements of this section are only relevant where Removals or Reductions are not the main purpose and the only source of revenue of a Project.

In such a case, evidence of Financial Additionality must be provided via full Project financials.

Proof of Financial Additionality requires, at minimum, the calculation of the return on investment via calculation of the Internal Rate of Return (IRR) for the Project.

The IRR determination must be completed in accordance with generally-accepted accounting practices. Furthermore,

the assumptions, data and conclusions of the analysis must be consistent with information presented to the company's decision-making management and investors/lenders. The analysis should be completed for a 10-year period, or if another length of time is chosen, justification must be given based on project circumstances. Non-depreciated residual values of any equipment or assets must be included as positive cash flows in the final year.

Existing government subsidies, public financing or tax incentives received must be taken into account as a revenue stream.

The Project Proponent must determine the IRR for the Project without Carbon Finance revenues. For new Projects, the Project must ultimately demonstrate:

- a zero or negative IRR; or
- an IRR of less than the cost of capital (required rate of return) or return on equity for the Project.

A Project may justify a higher IRR as additional if the Project Proponent can demonstrate that the Project requires such an IRR to enable financing for higher risk Projects, which may require higher interest rate loans, high interest rate venture or equity investment, or similar.

The Project must demonstrate that the economic performance of the mitigation activity increases decisively through carbon credit revenues, and that carbon credit revenues can raise the economic performance at or above the required financial benchmark (IRR).

The IRR analysis must also include a scenario analysis that demonstrates the ability to meet the above Additionality criteria for cases where values in the analysis change. Scenario analyses must include:

- variation of initial investment costs, if greater than 20% of total Project costs;
- variation of projected revenues to consider market changes for products or fees charged;
- evaluation of any assumption, value, or projection that accounts for greater than 20% of Project costs or Project revenues, or any value that impacts less than 20% of costs or revenues but which the Project Proponent determines can have significant impact (>10%) on IRR.

At a minimum, the IRR analysis must include variation of the factors above by ±20% or by a more appropriate value based on historical data or literature.

For pre-existing Projects:

- Projects must also provide an estimate of the quantity of Removals and/or Reductions that would occur in the absence of Carbon Finance.
- Removals and/or Reductions only in excess of this quantity may be considered Financially Additional.

3.5.3 Regulatory and Policy Additionality Considerations

Project Proponents must demonstrate Regulatory Additionality, meaning that the Project is not required by existing laws, regulations, policies, or other binding obligations. If the activity is required by such legal requirements, the Project is not considered Additional.

For UNFCCC non-Annex I countries[7], Project Proponents must only demonstrate that the Project is not required by any

systematically enforced laws, regulations, policies, or other binding obligations. In other words, the Project must not be required under “de facto” regulation, as opposed to “de jure” regulation. If relevant laws or regulations exist but are not systematically enforced, Project Proponents must provide evidence to demonstrate this lack of enforcement.

Legal requirements that must be considered include, but are not limited to:

- any national, state, municipality or other local law or regulation; and
- any policy, such as a procurement practice that requires the specific Project activity.

If the Project is legally required as described above, but the Project provides Removals or Reductions that exceed the legal requirement, the Removals in excess of the legal requirement may be considered additional if the other Additionality requirements are also met.

3.6 Leakage

Projects must demonstrate a robust assessment of potential increases in GHG emissions outside the defined GHG system boundary that occurs as a result of the Project activity.

Leakage may be a result of activity shifting, market transformation or shifts in ecosystem processes and can relate to both upstream and downstream activities.

Where the potential for such Leakage is identified, it must be quantified and deducted from the Removals or Reductions in accordance with the relevant Protocol.

3.7

Uncertainty in Removals and Reductions

Projects must report a list of all parameters that are used in the Removal or Reduction calculations.

A sensitivity analysis that demonstrates the impact of each input parameter's uncertainty on the final net CO_2e uncertainty must be provided alongside a GHG Statement submitted for verification.

Uncertainty information will vary depending on the data, but may include the type of distribution and parameters needed to describe uncertainty and/or shape of the data when plotted. Appendix A provides additional guidelines for assessing different types of uncertainties.

Furthermore, the source for information relevant to Uncertainty calculation and/or justification is required for each input parameter.

Details of the sensitivity analysis method must be provided so that the results can be re-created.

The stepwise process for reducing and documenting uncertainty is set out in Appendix B.

The Uncertainty information may be obtained from instrument calibrations, measurement variability, published literature, Proxies, expert judgment, and/or other Reputable Sources.

3.7.1

Conservative Estimate of Removals and Reductions

Projects must follow a procedure for incorporating Uncertainties into a Conservative estimate of Removal or Reduction. The following approaches are acceptable, and

other methods may be considered after an internal Isometric review.

A. Conservative estimate of input parameters

B. Variance propagation

C. Monte Carlo Simulations

Option A is suitable for Projects where it is difficult to obtain detailed information on input parameter distributions. Using conservative estimates of input parameters will lead to a Conservative Removal or Reduction estimate.

Option B can be used when parameters follow normal distributions and errors are linear (i.e., errors increase linearly as parameter values increase) and independent (i.e., changing one variable has no effect on another variable).

Option C provides a more comprehensive representation of Uncertainties and may result in more Credits Issued. This approach is suitable for Projects where sufficient information is available for all input variables so that Monte Carlo Simulations can be conducted.

As Removal and Reduction quantifications become better-constrained and more data is gathered, Protocols may progress from using Option A to Option B to Option C.

3.7.2

Guidance on Conservative Estimate of Input Parameters

For Option A, a uniform distribution is assumed for all parameters, unless it can be demonstrated otherwise. Isometric recommends that the value used in the Removal or Reduction calculation be either $\leq 16^{th}$ or $\geq 84^{th}$ percentile, depending on which one yields a more Conservative

estimate of Removal or Reduction. There may be instances where a parameter might only have a few reported values, and it does not make sense to assume a continuous distribution. In these cases, it would be more appropriate to use the minimum or maximum value as the Conservative estimate. See Appendix A for some examples.

3.7.3
Guidance on Variance Propagation

For Option B, the variance must be defined for all input parameters. Variance propagation should be conducted following uncertainty propagation rules[8], where the outcome is the variance in the Removal or Reduction. Isometric recommends that the Conservative Removal or Reduction estimate be at least 1 standard deviation (square-root of the variance) below the mean, equivalent to ≤16th percentile.

3.7.4
Guidance on Monte Carlo Simulations

For Option C, the input distributions must be specified for all input parameters. The final distribution for the Removal or Reduction is determined by randomly sampling from input distributions many times (e.g., n = 1,000).[9] The Conservative estimate of Removal or Reduction will be ≤16th percentile for consistency between Option B and C in the case of normal distributions.

3.7.5
Expert Reviews and Updates

Parameter Uncertainties that are based on expert judgment and which contribute the most significantly to the final Removal or Reduction calculation (based on a Sensitivity Analysis) will be subject to expert review by Isometric and/or selected Science Network experts.

3.8

Stakeholder Input Process

Relevant stakeholders must be informed of the Project's proposed and current activities by the Project Proponent.

All stakeholders must be equitably represented, involved and able to contribute freely. This includes but is not limited to Indigenous Peoples and Local Communities (IPLCs), stakeholders with land-tenure rights, local policymakers. Additional stakeholders may include regional or national government officials, local NGOs, or other groups likely to be affected by the project.

Consultation with stakeholders and rights-holders must be conducted and evidence of these meetings must be submitted to Isometric.

Consultations must meet the requirements below.

- Iterative:
  - The first consultation meeting must occur prior to Project development, such that any input and concerns can be incorporated into the Project's design, and meeting and correspondence must be operational throughout the Project's lifespan.
- Accessible:
  - Stakeholders and rights-holders must be invited to consultation meetings with a minimum notice of 14 days.
  - Stakeholders and rights-holders must be invited to consultation meetings via appropriate methods, which may include, but are not limited to, the post, email, or notices in newspapers and public places.
  - Consultation meetings must be scheduled to maximize attendance, taking note of cultural or

religious holidays and heritage.
    - Meetings, documentation and correspondence must be in the local language(s), or have a translator where necessary to facilitate communication.
    - Meetings, documentation and correspondence should be respectful of local knowledge; and accessible to a non-technical audience.
- Transparent:
    - The intention of each consultation meeting must be communicated to all stakeholders prior to the meeting.
    - All stakeholder or Project Proponent conflicts of interests must be declared.
- Free from external manipulation.

- Include a system for stakeholders to voice, process, and resolve grievances:

- Project Proponent contact information must be made available to all stakeholders.

- Grievances must be acknowledged no later than 14 days after receipt by the Project Proponent.

- Grievances must be resolved or escalated no later than 60 days after receipt.

- Systematically documented and made public or accessible upon reasonable request.

Project Proponents must inform Isometric about any new grievances that are raised no later than 14 days after receipt

by the Project Proponent.

3.9
Regulatory Compliance

Projects must identify and demonstrate within the PDD the method(s) for compliance with regulations for all jurisdictions to which the Project is beholden.

3.10
Environmental and Social Impacts

Projects must comply with national and local laws and regulations and, where relevant, international conventions and standards.

Projects must consider the material environmental and social impacts that could potentially arise as a result of their activities, both within and beyond its boundary, and at minimum must do no net environmental or social harm.

Remediation of any unintentional harm, caused directly or indirectly by a project, must be carried out by a Project Proponent. Failure to adequately remediate any harm caused may result in cessation of credit issuance. The consideration of these impacts must be ongoing throughout a Project's lifespan and assessments updated when necessary, and must include provisions for Project closure and post-closure.

For each aspect of the environmental and social impacts assessment, the Project must demonstrate how these risks have been assessed and, if applicable, what mitigation plan is in place to prevent them.

If risks are not applicable to the project this finding must be reported to Isometric, and environmental regulators where

applicable, and justified within the PDD.

A full Environmental and/or Social Impact Assessment (EIA and/or SIA) conducted by a third party is suggested for all Projects but is only required if impacts are considered significant and/or if required by the host jurisdiction.

It is the responsibility of the Project to ensure that any EIA or SIA undertaken meets all local, regional and national and international regulations and laws.

3.10.1 Environmental Impacts

Assessments undertaken to evaluate the potential for environmental harm by a Project may be undertaken with varying formats, names and costs, depending on the location of a Project. These assessments are commonly named environmental impact assessments (EIA) and are often required under specific environmental laws or regulations. Within some countries, such as the U.S., multiple levels or types of EIA may exist. Within the U.S. EIAs exist as Environmental Impact Statements (EIS) and Environmental Assessments (EA), with requirements for each type of assessment varying on a federal and state by state basis. It is the responsibility of the Project Proponent to identify specific requirements for EIAs within the location and type of a Project. Within this Standard all such assessments are referred to as Environmental Impact Assessments (EIA).

A Project must demonstrate that it creates no net environmental harm.

These assessments and mitigation strategies must consider the potential negative environmental risks from a Project's implementation. This must include, but is not limited to, the following, as outlined in the ICVCM Core Carbon Principles[10]:

1. Resource efficiency and pollution prevention, including:
2. Pollutant emissions to air.
3. Pollutant discharges to water, noise and vibration.
4. Generation of waste and release of hazardous materials, chemical pesticides and fertilizers.

2. Biodiversity conservation and sustainable management of living natural resources:
3. Avoid, or where this is not feasible, minimize negative impacts on terrestrial and marine biodiversity and ecosystems.
4. Protect the habitats of rare, threatened, and endangered species, including areas needed for habitat connectivity.
5. Do not convert natural forests, grasslands, wetlands, or high conservation value habitats.
6. Minimize soil degradation and soil erosion.
7. Minimize water consumption and water stress in any necessary mitigation activities.

3.10.2 Social Impacts

A Project must demonstrate that it creates no net social harm.

These assessments and mitigation strategies must consider the potential negative social risks from a Project's implementation. This must include, but is not limited to, the following, as outlined in the ICVCM Core Carbon Principles[10]:

1. Labor rights and working conditions including:

2. Providing safe and healthy working conditions for employees. Health and Safety risk assessments and mitigation strategies in accordance with local, national and international laws and guidelines are required by every Project for each stage of their operations and must ensure the nature of the work being undertaken is captured.
3. Providing fair treatment and equal opportunities to all employees, including provision of equal pay for equal work, regardless of age, gender, race, ethnicity, religion, disability, sexual orientation, sexual identity, education, national origin, or any other distinguishing characteristic or trait.
4. Prohibiting the use of forced labor, child labor or labor by trafficked persons.
5. Protecting third party contracted personnel.

2. Land acquisition and involuntary resettlement, including:
3. Actively avoiding physical or economic displacement.
4. Where not possible, minimizing the amount and the impact.

3. Impacts on Indigenous People (IP), Local Communities (LC) and cultural heritage; The project must:
4. Identify any direct or indirect impacts on IPLCs, including customary rights.
5. Recognise, respect and promote the protection of the IPLCs rights in line with applicable international human rights laws and UN Declarations, including International Labour Organisation Convention 169 on Indigenous and Tribal Peoples[11] and the United Nations Declaration on the Rights of Indigenous Peoples[12].
6. Preserve and protect cultural heritage and ancestral knowledge aligned with IPLCs and any UNESCO Cultural Heritage Conventions.

7. Not force eviction or physical or economic displacement.
8. If there are any potential negative impacts[13] or displacement is required, the Project must not proceed without the free and prior and informed consent (FPIC)[14] of the relevant IPLCs[15].

4. Respect for human rights and stakeholder engagement, including:
5. A Stakeholder Input Process is in place and aligns with the guidance outlined in Section 3.8 of the Isometric Standard.
6. Identifying, avoiding, mitigating and remediating any human rights impacts (as defined by international treaties, organizations, and conventions including the United Nations International Bill of Human Rights[16]).
7. Protecting against gender-based violence against women and girls, including sexual harassment, human trafficking, slavery, imprisonment, physical and mental drudgery, and punishment or coercion.

3.10.3 Sustainable Development Impacts

Projects must demonstrate, where relevant and feasible, how their carbon removal activities are consistent with relevant Sustainable Development Goals (SDG) objectives of all jurisdictions in which they operate.

The SDGs are a set of 17 global and interconnected goals established by the United Nations in 2015 to address a wide range of social, economic and environmental issues.

Such an explanation must be included in the Project Design Document. If applicable, a qualitative assessment must be included for any positive impacts identified in relation to

SDGs other than SDG13. Project Proponents must provide information on any standardized assessment tools and methods used as part of this explanation.

3.11
GHG Statement Policies

Projects credited under the Isometric Standard must use project-level consequential analysis to determine net removals or reductions associated with project activities. Projects must be assessed against a baseline scenario of the project not taking place. Attributional analysis is acceptable as a proxy for project emissions accounting.

Projects may submit Removals and Reductions in the same GHG Statement but they must be reported separately:

- The Project's removals, associated counterfactual and emissions allocated to removals must be presented together in net metric tonnes of CO_2e.
- The Project's reductions, associated counterfactual and emissions allocated to reductions must be presented together in net metric tonnes of CO_2e. Gross reductions and associated counterfactuals must be subdivided into avoided fossil CO_2 and avoided non-CO_2 greenhouse gasses.

All Projects must follow GHG accounting requirements as set out by the relevant Protocol and Modules.

This includes following GHG Statement boundaries and adhering to GHG accounting principles such as data quality and materiality.

3.12
Data sharing

All evidence and data related to the underlying quantification of Removals or Reductions will be available to the public through Isometric's platform for any Verified Credit. This includes:

- Project Design Document
- Measurements taken
- Emission factors used
- Scientific literature used

The Project Proponent may request for confidential information to be restricted, whereby it will only be available to authorized Buyers, Isometric and designated VVBs (during Validation or Verification). This includes emissions factors from licensed databases. However, all other numerical data produced or used as part of the quantification of CO_2e removed or reduced may not be restricted in this way.

4.0

Validation and Verification Requirements

This section defines the requirements for Validation and Verification of Projects on the Isometric Registry. Accredited VVBs Validate Projects and Verify Removals and Reductions in line with this Isometric Standard and a Certified Protocol.

4.1

Validation and Verification Body Qualification Requirements

All VVBs are approved by Isometric independently and impartially based on alignment with Conflict of Interest policies, rotation of VVB policies, oversight on quality, and the following requirements:

- VVBs must be able to demonstrate accreditation from:
  - an International Accreditation Forum[17] member against ISO 14065 or other relevant ISO standard, including but not limited to ISO 14034, ISO 17020, ISO 17029; or
  - a relevant governmental or intergovernmental regulatory body.
- Alternatively, on a case-by-case basis, if VVBs are able to demonstrate to Isometric that they satisfy all required Verification needs and competencies required for the relevant Protocol and follow the guidelines of ISO 19011 or other relevant standards, they may be approved.

4.2

Validation and Verification Process

All Projects must undergo an initial Project Validation, carried out by an independent VVB, to assess conformity with the Isometric Standard, and with the applied Certified Protocol. Once a Project has been Validated, the Project is subsequently able to submit GHG Statements to Isometric. Isometric requires Verification of any GHG Statements in order to Issue Credits. Validation and Verification will be carried out in line with ISO 14064-3 and ISO 14065. Additional requirements are listed below.

- Unless otherwise specified in the applicable Protocol or Module, Removals or Reductions may be Verified annually, or more frequently, according to the requirements of the relevant Certified Protocol, or at the Project Proponent's request.
- Project Validation may occur at the same time as the first Verification, or before the first Verification.
- If a Project Proponent wishes to renew the Crediting Period of a Project, an updated PDD must be provided and the Project must be re-Validated.
- Unless otherwise specified in the relevant Certified Protocol, a site visit is required during the first Validation or Verification of a Project. For subsequent Verifications, the VVB must identify whether a site visit is needed, based on an independent risk assessment.
- The level of assurance for Verifications must be reasonable.

Prior to Validation/Verification, the VVB must prepare a Validation/Verification plan that details the activities and schedules. The plan may be revised as necessary during the

process. The plan must be communicated with the Project Proponent and must include, at minimum:

- the scope and objectives;
- identification of the Validation/Verification team and their roles;
- client/responsible party contact;
- schedule of activities;
- level of assurance;
- Verification criteria;
- Materiality; and
- schedule for site visits.

The Verifier must produce a report that documents the activities, results, findings, and conclusion of the Verification. The Verification report must contain, at a minimum[18]:

- a statement that the Project Proponent is responsible for the fair presentation of the PDD in accordance with the criteria;
- a statement that the Verifier is responsible for expressing an opinion on the PDD based on the Verification;
- a description of the Verification evidence-gathering procedures used to assess the PDD;
- the Verification opinion, which will will state the net Removal and/or net Reduction during the covered period to a reasonable Level of Assurance;
- the date of the report;
- the Verifier's location;
- the Verifier's signature;
- a summary of the GHG Statement;
- reference to the Verification criteria; and
- Verification scope.

Following this process, Isometric will review the Project documents used for Validation and Verification, and either accept them, request corrections or clarifications for the Project Proponent or VVB to address, or reject them. Once Isometric has reviewed and approved the initial Validation Report for a Project, the Project will be listed as Validated on the Registry, with the date of Isometric's approval listed as the "Approval Date" (also known as a "registration date") of the Project. For Verifications, once Isometric has accepted a Verification Report, the corresponding Removals and Reductions will be deemed Verified, and eligible for issuance of Credits.

For both Validations and Verifications, the results of the Validation/Verification process, including the Validation/Verification Report and Opinion, will be made available for each Project.

4.3 Materiality Threshold

The threshold for Materiality, considering the totality of all omissions, errors and mis-statements, is 5% of net Removal or Reduction for all Projects. To accept a Verification Opinion, Isometric requires that discrepancies between the Removal or Reduction Claimed by the Project Proponent and that estimated by the VVB for a given statement be less than the Materiality threshold.

Qualitative materiality issues may also be identified and documented, such as[19]:

- control issues that erode the verifier's confidence in the reported data;
- poorly managed documented information;
- difficulty in locating requested information;

- noncompliance with regulations indirectly related to GHG emissions, removals or storage.

4.4 Conflicts of Interest

Any organization which has been involved in the development of a particular Project may not act as a VVB for Validation and/or Verification purposes for that Project.

To minimize the risk of conflicts of interest occurring between the Project Proponent and the VVB, Isometric will select and engage VVBs for Project Validation and Verification, and VVBs must complete a conflict of interest disclosure.

4.5 Rotation of Validation and Verification Bodies

Isometric requires that Projects must work with a single VVB for no longer than five consecutive years. A given VVB may conduct Verification for a Project during no more than five out of seven consecutive years.

4.6 Validation and Verification Body Oversight

In addition to the qualification requirements, Isometric may oversee VVB activity during the Validation and/or Verification process, and may suspend approval of a previously approved VVB. Oversight can include review of VVB documentation, including Verification and sampling plans, reports, opinions and conflict of interest disclosures, as well as review of Project Proponent documentation.

Isometric will report significant and/or repeated VVB performance concerns to the relevant accreditation body.

5.0

Crediting

Isometric issues two Credit Types:

- Carbon Dioxide Removal credits, representing one net metric tonne of verified CO_2e removal, and
- Emission Reduction credits, representing one net metric tonne of verified CO_2e reduction of specified greenhouse gasses generated by an existing emitter.

Credits may be Issued, Transferred and Retired on the Isometric Registry according to the rules and requirements set out in this section.

Isometric is the sole Credit issuing body on the Isometric Registry.

5.1

Credit Attributes

Credits have only one owner at any time. A full ownership history is tracked and is publicly visible for each Credit. Each Credit is Issued Ex-post against a net Verified Removal or Reduction thereby ensuring every net tonne of CO_2e emission impact is Credited and accounted for only once, and that any Credit can be traced back to an individual Removal or Reduction.

Each Credit has a publicly available history of ownership that indicates when the Credit was first Issued, who it was first Issued to, if and when it was subsequently Delivered or Transferred, if and when it was Retired and by whom, on behalf of whom.

When first Issued, a Credit's status is Active. Upon Retirement of a Credit, its status becomes Retired. Similarly, upon Cancellation of a Credit (for example, in compensation for erroneous overissuance, or in compensation for a Reversal as outlined in Reversals and Buffer Pools), its status becomes Canceled.

Credit metadata include:

- Unique Serial Number;
- Type (Removal or Reduction);
- Durability;
- Protocol version the Credit was issued against;
- Issuance Date;
- Issuing Project;
- Issuing Project Proponent;
- Country of Removal or Reduction;
- Ownership history, including the current Owner (the Owner who is retiring the Credit) and all previous Owners and Transfer Dates;
- Retirement Date;
- Retirement Beneficiary;
- Credit Status; and
- Vintage.

5.2
Issuance Process

Credits are Issued representing a specific Removal or Reduction. This process is undertaken once the net Removal or Reduction has been Verified against the respective Protocol. Credits are always Issued to the Project Proponent of the Removal or Reduction. The Project Proponent is then able to Deliver Credits to Buyer Credit Accounts on the

Registry, and can do so at a time and in an order of their own discretion.

5.3

Transfer and Delivery Rules and Ownership History

Deliveries can be made by Project Proponents to Buyer Credit Accounts, resulting in transfer of ownership of one or more newly Issued Credits. Further Transfers can be made by Credit Account holders to other Credit Accounts, resulting in transfer of ownership of one or more Credits. For each Credit, the initial Delivery and all Transfers are logged on the Credit and made publicly available in the Credit's history.

Project Proponent Deliveries are subject to the following rules:

- Project Proponents can only Deliver Credits that have been Issued;
- Project Proponents can only Deliver Credits that they own; and
- Project Proponents can only Deliver Credits to Buyers who have an active Isometric Registry Credit Account.

Buyer Transfers are subject to the following rules:

- Buyers can only transfer Credits that they own; and
- Buyers can only transfer Credits to another organization that has a Credit Account on the Isometric Registry.

5.4

Retirement Rules

Retirement is the mechanism by which a Credit's ownership state is finalized. This ensures that once the net tonne of CO_2e represented by the retired Credit is used towards an accounting activity, it can never be used again by the owner, the Beneficiary of the retirement, or any other actor.

The Beneficiary of a Retirement is the organization on behalf of whom the Credit was retired, and must be publicly identified. Beneficiaries can be the current holder of a Credit at the time of retirement, or an organization that is specified by the Credit Account holder during the Retirement procedure. The Credit Account holder can retire Credits by following the Retirement process through their Credit Account on the Registry.

The owner of an unretired Credit may publicly market its ownership of said Credit, as well as the potential for the Credit to be retired by any current or future owners, but only the Beneficiary of a Retired Credit may claim the environmental benefits associated with that Credit.

Retirement is subject to the following rules:

- a Credit Account holder can only retire Credits they own, which are in their Credit Account;
- a Credit can only be Retired once; and
- any number of Credits can be retired at any given time, provided that number of Credits does not exceed the number of Credits held by the Credit Account.

5.5

Retirement Certificates

Retirement Certificates serve as proof of the Retirement of one or more Credits, and are produced only on completion of a Retirement. Every Retirement Certificate relates to one or more Retired Credits. If more than one Credit is Retired at a time, the Certificate will list all Retired Credits. Retirement Certificates show the relationship between retired Credits and the Removals or Reductions they derive from.

5.6 Reversals and Buffer Pools

This section outlines how Reversals are handled on the Isometric Registry, through the use of Buffer Pools.

All Reversals will be reviewed by the Isometric Science Team.

Note that changes to calculations as a result of updates to Protocols are not considered Reversals, and will generally not affect previously issued Credits. However, Isometric will monitor for significant impacts due to Protocol updates to ensure overall quality and to address system level impacts (e.g. considerable evolution of scientific consensus).

5.6.1 Reversals

Project Proponents must monitor for Reversals as prescribed by a Project's Monitoring Plan, and promptly report potential Reversals to Isometric if identified. Isometric's expectation is for such a report to be made within one business day of having been identified, but must be made within a maximum of three business days. At the verification following a Reversal, the Project Proponent must report relevant monitoring data for any Reversal which has occurred, to be assessed by the verifier as part of that

verification. Reversal data from Projects will be made public.

When a Reversal is identified, Credits will be Canceled from the Project Proponent's Buffer Pool to compensate for the Reversal, according to the requirements in Buffer Pools.

In the event that a Project Proponent is temporarily unable to comply with monitoring requirements as outlined in the relevant Protocol, the treatment of possible Reversals will be assessed on a case by case basis by Isometric, in consultation with the Project VVB.

Furthermore, Reversals are classed as either Avoidable or Unavoidable, according to the following requirements:

- Where a Project Proponent ceases ongoing monitoring, this will be deemed an Avoidable Reversal.
- Where a Reversal occurs over which the Project Proponent has influence or control, including for example in relation to poor project management or operational negligence, this will be deemed an Avoidable Reversal.
- Where a Reversal occurs over which the Project Proponent has no control, including natural disasters, encroachment by outside actors, or occurrence of other demonstrably unforeseeable factors reasonably considered to be outside of the Project Proponent's control, this will be deemed an Unavoidable Reversal.

The classification of a Reversal as either Avoidable or Unavoidable will be made by Isometric, in consultation with the Project VVB, according to the above requirements. This classification has implications in relation to the Project Proponent's Buffer Pool, as outlined in Buffer Pools.

5.6.2

Buffer Pools

Isometric will maintain a Buffer Pool of Credits in accounts specific to each Project Proponent, from which Credits will be Canceled in the case of a Reversal. The number of Credits to be Canceled must equal the size of the Reversal.

In the case of Biosphere Protocols, a Buffer Pool of Credits specific to each Protocol will be maintained, and further requirements applicable to such Buffer Pools are set out in the applicable Protocols.

Isometric reserves the right to amend its Buffer Pool approach given notice and consultation with Project Proponents in line with this Standard's Updates to Protocols policy.

By default, all Projects must make buffer pool contributions, except for Projects which meet either of the following criteria:

1. Projects deemed to have a “No observable risk” Risk of Reversal, as indicated in 2.5.9 Risk of Reversal;
2. Projects operating within a recognised regulatory scheme that requires compensation for reversals, as outlined in 5.6.2.1 Recognition of Equivalent Regulatory Schemes.

For Projects contributing to a buffer pool, whenever Credits are Issued following a Verified Removal or Reduction, a percentage of these Credits will be Issued to the Project Proponent's Buffer Pool account, according to the Buffer Pool Size below.

Credits in a Buffer Pool cannot be Transferred, or used to make Deliveries during the Crediting Period or after the Crediting Period has ended, as they are held to accommodate for as long as there is a risk of Reversal from a Project. Exceptions may occur when Uncertainty decreases and thus Buffer Pool size is amended for a Project Proponent, in which case extra Buffer Pool Credits may be used to make Deliveries.

In the case that there are insufficient Credits in a Project Proponent's Buffer Pool to compensate for any triggered Reversals, all further Credits Issued from Removals and/or Reductions conducted by the Project Proponent (whether from the same Project, or other Projects run by the same Project Proponent) will be assigned to their Buffer Pool and Canceled, until all such Reversals have been fully compensated. Cases where Project Proponents cease operations will be addressed on an individual basis.

The Buffer Pool Size is determined by the Risk of Reversal of a given approach, as outlined below. The Risk of Reversal is defined in the relevant Certified Protocol. Projects with a higher Risk of Reversal have a correspondingly larger Buffer Pool. Note that the Buffer Pool is applied in addition to the Conservative approach taken in the quantification of net CO_2e Removal or Reduction, whereby uncertainty is factored into the Credited Removal or Reduction. The following Buffer Pool sizes may be subject to change in future Standard versions based on empirical data.

RISK OF REVERSAL	BUFFER POOL SIZE
No observable risk	0%
Very low	1%
Low	5%
Medium	7%
High	10-20%

Where an Avoidable Reversal has occurred, a Project Proponent must reimburse their Buffer Pool account by transferring a number of Credits equal to the size of the Reversal to the Buffer Account. While there are any outstanding Avoidable Reversals to be remedied in this way, all further Credits Issued from Removals and/or Reductions conducted by the Project Proponent (whether from the same Project, or other Projects run by the same Project Proponent) will be assigned to their Buffer Pool, until all such Reversals have been fully remedied.

Where an Unavoidable Reversal has occurred, the Project Proponent is not required to reimburse their Buffer Pool account.

5.6.3
Buffer Pool Refunds

Where a Project's Risk of Reversal assessment is revised downward, or where Isometric reduces the buffer pool contribution rate associated with a given risk category, the Project Proponent may be entitled to request a refund of the difference between credits previously contributed to the buffer pool and the amount that would have been required under the revised assessment or rate, so long as no Reversals have been triggered by the Project.

Any such refund shall be contingent upon confirmation of the revised risk of reversal assessment by the Project’s VVB at the Project's next verification. Upon successful verification confirming the lower risk assessment, and confirmation of a refund request by Isometric, the applicable number of Credits will be transferred to the Project Proponent’s account.

For the avoidance of doubt, where a change in Risk of Reversal assessment results from a material change in

storage technology or a change to the Project's selected Storage Module(s) applicable only to new issuances—such that previous issuances would continue to be assessed at the original Risk of Reversal level—refunds under this section shall not apply to those previous issuances.

5.6.4 Recognition of Equivalent Regulatory Schemes

Where a Project Proponent can demonstrate that a Project is operating within a regulatory scheme that requires compensation for reversals, the Project may forgo Buffer Pool contributions. Recognition of an equivalent regulatory scheme means that reversals occurring under that scheme are considered compensated through the scheme's own mechanisms, and the Project Proponent is not required to compensate for reversals through the Buffer Pool or other procedures defined in this Standard. The Project Proponent remains responsible for monitoring and reporting reversals in accordance with the Reversals requirements of this Standard.

The requirements for regulatory schemes to be considered as equivalent are:

- A binding cap-and-trade scheme with zero free allowances for storage operators, such that any reversal is sufficiently compensated by an equivalent reduction in GHGs under the scheme.
- A legally mandated requirement to be able to cover the full financial liability of the maximum possible storage inventory.

The following regulatory schemes are recognized as providing qualifying coverage:

- **EU CCS Directive (Directive 2009/31/EC)**: Projects storing CO_2 in geological formations within EU member states, where the storage operator maintains financial security in accordance with Article 19.

Suppliers seeking recognition under other regulatory schemes should contact Isometric with details of the applicable requirements. Recognition will be granted where Isometric determines the scheme provides substantially similar or improved protection compared to the buffer pool mechanism.

5.6.5

Future Recognition of Insurance for Reversal Risk Mitigation

Isometric is exploring the recognition of insurance as an alternative mechanism for mitigating reversal risk. Where Project Proponents are able to demonstrate qualifying insurance coverage that provides equivalent or improved protection against reversal events, Isometric may in future permit reduced or replaced buffer pool contributions. Criteria for qualifying insurance arrangements will be published as they are developed.

Project Proponents interested in exploring insurance as an alternative to buffer pool contributions are encouraged to contact Isometric.

5.7

No Double Counting

Isometric has rules and procedures in place to mitigate the risk of occurrence of Double Counting, which can occur in different ways, including double issuance, double use and double claiming.

Double issuance (occurring when more than one unique unit is Issued for the same Removal or Reduction activity) is not permissible, and Isometric has checks in place to mitigate the possibility of double issuance. Any Removal or Reduction activity listed on the Isometric registry must not be listed on another program, and may not be used to make a separate Removal or Reduction claim elsewhere. If the Project is associated with production of a commodity which is claiming an environmental attribute, the removals, or reductions may not be counted towards this environmental attribute.

In addition, it is not permissible to issue Credits for Removals or Reductions resulting from Removal or Reduction activities where credits related to the same climate impacts of the activity are traded in any other carbon crediting program.

Furthermore, it is not permitted to register Removal or Reduction activities or Projects with overlapping GHG accounting boundaries, except in the following circumstances:

- Projects whose physical Removal activities incorporate carbon sequestration which would individually be eligible under separate Isometric Protocols, and which may be registered as individual Projects under this Standard using separate Protocols; or
- Projects which incorporate both Removal and Reduction activities that would individually be eligible under separate Isometric Protocols, and which may be registered as individual Projects under this Standard using separate Protocols.

In both of the above circumstances, the GHG accounting boundary of the Project must not overlap with activities

registered with any other carbon crediting program. Secondly, the GHG accounting boundary must be clearly described between such Projects to ensure that all carbon fluxes are fully accounted for precisely once, thereby avoiding the possibility of double issuance.

Double use (occurring when an Issued credit is further transferred, used or retired after already having been retired) is mitigated through a transparent, public record of all Credits and Retirements, ensuring all Retirements used as claims towards mitigation targets can be uniquely identified, and can be traced back to the specific Removal or Reduction activity the Credit was Issued against. Once a Credit has been Retired, it cannot be used further, and a unique, public Retirement Certificate is produced.

Double claiming (occurring when an Issued credit is claimed twice towards achieving mitigation targets or goals) is not permitted on the Isometric Registry. For any Credit, no separate Removal or Reduction claims may be made for the underlying Removal or Reduction from which the Credit was Issued. Isometric will monitor for instances of Double claiming, and Credit Account users engaging in double claiming may have their Credit Account suspended.

5.7.1 Prevention of Scheme Hopping

To mitigate the risk of Double Counting and to prevent Projects seeking to avoid undertaking verification, monitoring, or corrective requirements through registration under alternative standards or programmes, Projects undergoing Validation must disclose:

- Whether the Project Proponent (or its legal predecessor) is currently participating in, or has

participated in, another carbon certification scheme, carbon registry or standard within the last 5 years.

- Audit reports from the last two Validation and Verification processes under any previous scheme. This must include a detailed list of findings and any history of suspension or withdrawal in the last 5 years.
- Whether the Project Proponent withdrew from a previous scheme prior to the completion of a first Verification.

5.8 CORSIA

Project Proponents wishing to issue credits that would be eligible as emissions units for the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) must comply with Isometric's CORSIA Eligibility Policy.

To mitigate the risk of double claiming of any CORSIA-eligible emissions units, Project Proponents must agree with Isometric suitable mechanisms to compensate for double claims of their CORSIA-eligible emissions units by airline operators for CORSIA and Host Countries working towards fulfillment of Nationally Determined Contributions (NDCs).

Project Proponents of CORSIA-eligible emissions must also comply with the information and reporting requirements in Isometric's CORSIA Eligibility Policy, including monitoring relevant Host Country reporting and submitting accurate, timely information on corresponding adjustments to Isometric.

Project Proponents must report their sustainable development contributions or co-benefits in their Project Design Document, including indicators specified by Isometric

and those that are publicly specified by a host country, for Credits to be identified as CORSIA Eligible Emission Units.

For Projects wishing to issue CORSIA Eligible Emission Units, a CORSIA Reversal Buffer Pool of Credits will be maintained with contributions determined by the Risk of Reversal defined in the relevant certified Protocol. More detail can be found in Isometric's CORSIA Eligibility Policy.

Isometric commits to being the final guarantor of CORSIA-eligible emissions units, if other safeguards to reimburse CORSIA-eligible emissions units, such as a project's buffer pool contributions, other mechanisms outlined in Buffer Pools or insurance policies, are insufficient to fully compensate for any reversal of issued CORSIA-eligible emission units. Further detail is provided in the CORSIA Eligibility Policy.

5.9 Non-Conformities and Sanctions

5.9.1 Non-Conformities

During a Validation or Verification conducted by a VVB, if any non-conformities with Protocol or Standard requirements are identified, the VVB will classify these non-conformities according to three categories: Minor Non-Conformity, Major Non-Conformity, and Critical Non-Conformity.

The categorization of the non-conformity will determine the severity of the associated sanction Isometric will apply to the Project Proponent.

- Minor Non-Conformity: a violation of the Isometric Standard or Protocol that has a limited impact,

constitutes an isolated or temporary lapse, and would not result in a fundamental systemic failure if not corrected.

- Major Non-Conformity: a violation of the Isometric rules or procedures that has been repeated by the Project Proponent, revealing systematic problems either in combination with further non-conformities or on its own merit. This includes, but is not limited to:
  - systematic problems with GHG data reported, such as incorrect documentation identified in more than 10% of the claims included in the representative sample;
  - failure to provide relevant information to the VVB, such as the information needed for the purposes of an audit;
  - omission of a Project Proponent declaring its participation in other carbon removal or GHG reduction certification schemes during the certification process;
- Critical Non-Conformity: violations that jeopardize the integrity of the Isometric Standard, including fraud, irreversible non-conformity, falsification of GHG data, or deliberate misstatement of activity. For instance, fabrication of data or submission of fraudulent GHG statements.

For the purposes of the issuance of a CRCF certificate of compliance, a failure to meet the requirements of relevant Modules and Protocols approved under the CRCF, as well as the quality criteria set out in Articles 4 to 7 of Regulation (EU) 2024/3012[20], will also be categorized as a Critical Non-Conformity.

5.9.2 Sanctions

Based on the classification of the non-conformities in Non-Conformities, Isometric will apply the following sanctions, which are in accordance with the CRCF implementing regulation. The applicable sanction applied will depend on whether the Credits or CRCF Certificate of Compliance have been issued.

In the case of a Minor Non-Conformity, the Project Proponent must remediate the issue during the current Validation or Verification, unless the VVB determines that it may be recorded as a Forward Action Request. Any such Forward Action Request must be addressed at the next verification, or by a date specified by the Verifier, and in any event within 12 months at the latest.

In the case of a Major Non-Conformity, if a Project has not been validated or a CRCF Certificate of Compliance has not been issued, the project will not be validated and no CRCF Certificate of Compliance will be issued until the non-conformity has been resolved. Where a CRCF Certificate of Compliance has already been issued or the Project has been validated, its Certificate of Compliance and further credit issuance to the Project will be suspended. The Project Proponent must implement remedial measures within 90 days of the major non-conformity being identified. If this remedial action does not occur within 90 days, the Project's CRCF Certificate of Compliance will be cancelled and the Project will be delisted from the registry.

In the case of Critical Non-Conformity identified during a Verification or monitoring audit, or through Isometric's internal monitoring or complaints process, no further credits will be issued. Any Credits previously issued that are

attributable to the Non-Conformity will be deemed as having undergone a Reversal, and the procedure in Reversals will apply. Any CRCF certified units will be subject to immediate cancellation and the remediation procedures described in Remediation for Fraud will apply.

In the case of a CRCF certificate of compliance, the certificate will be cancelled and there will be no further issuance of CRCF certified units.

Any further credit issuance to the Project will be suspended. In the case of Projects which are not seeking eligibility under CRCF, the Project Proponent may re-apply to issue credits if they demonstrate that the issues which lead to the Critical Non-Conformity have been thoroughly addressed.

6.0

CRCF

Project Proponents wishing to issue credits under the EU's Carbon Removal and Carbon Farming certification scheme are subject to additional procedures and requirements to ensure they comply with the applicable legislation.

Projects seeking Validation under the Isometric Standard must indicate whether they wish to be approved to issue CRCF certified units. If so, Projects must use an appropriate Isometric Protocol and must comply with the procedures and requirements described in this Section.

6.1

Appointment of Certification Bodies

In addition to the Validation and Verification Body Qualification Requirements set out in Validation and Verification Body Qualification Requirements, Isometric will apply the following additional requirements when appointing a VVB as a Certification Body under the CRCF:

The VVB must have the relevant legal status to perform certification activities through one of the following qualifying criteria:

- accreditation by a National Accreditation Body in accordance with Regulation (EC) No 765/2008; or
- compliance with EN ISO/IEC 17065 and EN ISO/IEC 17029 or EN ISO 14065.

If the VVB is not accredited by a National Accreditation Body in accordance with Regulation (EC) No 765/2008, it must select and appoint the audit team in accordance with EN ISO/IEC 17021-1 in conjunction with EN ISO/IEC 19011,

taking into account the competence needed to achieve the objectives of the audit.

The VVB must also meet the following requirements:

- have the knowledge, experience, skills and capacity necessary for effectively conducting the audit, including a minimum of 2 years' experience in Life-Cycle Assessment, or proven relevant level of capacity;
- have specific experience in auditing GHG emission calculations in accordance with the relevant Protocols and Modules;
- depending on the specific types of activity audited, have additional experience as relevant in the fields of agriculture, agronomy, ecology, forestry, natural science, silviculture, engineering, building materials and embodied carbon, energy management or a related field;

Verification reports issued must include all information listed in Annex III of the Commission Implementing Regulation.

Isometric will run training courses for VVBs. This will include an exam which tests an auditor's (verifier's) overall technical expertise in the area. Auditors must participate in the training courses before performing audits for Isometric. Isometric will also provide guidance to VVBs on other relevant aspects of CRCF certification, including updates on the regulatory framework or findings from Isometric's internal monitoring process.

6.2
Exclusion Criteria

For Projects seeking eligibility under CRCF, Isometric will exclude a Project from this Standard in the following

circumstances:

- The Project Proponent (or its legal predecessor) failed to disclose the relevant information required under Prevention of Scheme Hopping.
- The Project Proponent (or its legal predecessor) failed a Verification under another scheme, unless said audit occurred more than 3 years prior to the current application or the previous certification scheme has ceased operations.
- The Project Proponent (or its legal predecessor) withdrew from another scheme prior to the first Verification, unless the Project Proponent can demonstrate the withdrawal was unavoidable and not caused by a Critical or Major Non-Conformity.

6.3 Grievances

Isometric has a CRCF compliance policy in place. This compliance policy allows for complaints and whistleblowing to Isometric with respect to VVBs and Project Proponents.

Isometric ensures the protection of natural or legal persons who report infringements or lodge complaints in good faith in accordance with the Whistleblower Protection Directive (EU) 2019/1937 of the European Parliament and of the Council.

Isometric will maintain a register of all complaints made which relate to Projects issued under the CRCF. Upon request from a Member State or the Commission, Isometric will share all documents related to the complaint and its handling.

As standard, Isometric will pursue the approach to remediation described in Isometric's CRCF compliance policy. Where necessary, Isometric will apply the relevant remediation measures and sanctions for non-conformity by Project Proponents as set out in Non-Conformities and Sanctions of this Standard. Where necessary, Isometric will take corrective measures on its governance structure or on its internal monitoring process.

6.4
Remediation for Fraud

In cases where CRCF certificates have been issued on the basis of fraudulent information, the associated Credit Account of the Project Proponent will be suspended and any necessary compensation will be provided through cancellation of certified units.

The Project Proponent must procure an equivalent number of CRCF units and immediately cancel them within 90 days of notification. If the Project Proponent fails to do this, as a last resort Isometric will procure a corresponding number of CRCF certificates and cancel them.

6.5
Internal Monitoring

Isometric will implement a system of internal monitoring to verify compliance of Project Proponents with the rules and procedures applied by this Standard and to ensure the quality of the work carried out by the auditors of the Certification Bodies.

Internal monitoring will be carried out at least once a year or at least at the same frequency of the audits to reflect the certification scope, as well as the level of risk of the activities carried out by the Project Proponents.

As part of the internal monitoring, Isometric will require Certification Bodies to provide all reports of the Validation (certification audit), Verification (re-certification audit), or monitoring audit. The internal monitoring will cover a random and risk-based sample of those audit reports by each Certification Body.

6.6 Planning and Reporting

This section details requirements from Annex I of Regulation (EU) 2024/3012.

6.6.1 Activity Plans

A CRCF activity plan is equivalent to an Isometric Project Design Document (PDD). PDD requirements are detailed in the Documentation section of the Isometric Standard. The following additional requirements apply to all Isometric CRCF PDDs:

- All entities in the carbon removal value chain involved in delivery of the activity must be identified as Project participants.
- For Removals, the PDD must demonstrate that estimates of $CR_{baseline} - CR_{total} - GHG_{associated} > 0$, where:
  - $CR_{baseline}$ is the amount of carbon removals under the baseline;
  - CR_{total} is the total amount of carbon removals of the activity;
  - $GHG_{associated}$ is the increase in direct and indirect greenhouse gas emissions over the entire lifecycle of the activity which are attributable to its implementation, including indirect land use change,

calculated, where applicable, in accordance with the protocols set forth in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories and any further refinement to these 2006 IPCC Guidelines.

- PDD must include a description of the materiality tests applied by the PP in accordance with the Materiality section of the Isometric GHG Accounting Module or the relevant Isometric CRCF Protocol.

6.6.2 Monitoring Plans

A CRCF monitoring plan is equivalent to the Monitoring Plan section of each Isometric Project Design Document (PDD), in conjunction with the Project's preliminary GHG Statement submitted during Validation.

All Isometric CRCF PDDs and GHG Statements must include:

- a description of the procedure for managing the competences of personnel responsible for monitoring and reporting where applicable, the default values used for calculation factors indicating the source of the factor, or the relevant source, from which the default factor will be retrieved periodically;
- where applicable, a list of laboratories engaged in carrying out relevant analytical procedures; where measurements are taken, a description of the measurement method including descriptions of all written procedures relevant for the measurement;
- application of the Quality Assurance requirements in Quality Assurance.

Project Proponents must keep records of all relevant data and information for at least 10 years after the end of the

Monitoring Period.

6.6.3
Monitoring Reports

A CRCF monitoring report is equivalent to the GHG Statement submitted during Verification of Credits.

A GHG Statement contains all parameters measured and calculated for the quantification of removals and emissions of GHGs associated with the Project. The contents of a GHG Statement are defined by the relevant Isometric CRCF Protocol.

6.7
Other Delegated Act Requirements

6.7.1
Global Warming Potential of non-CO2 GHGs

All Isometric CRCF Projects must convert emissions of non-CO2 GHGs to CO2e using the 100-year Global Warming Potentials set out in Annex I to the Delegated Regulation (EU) 2020/1044.

6.7.2
Quality Assurance

Project Proponents must:

- ensure that all relevant measuring equipment is calibrated, adjusted and checked at regular intervals, including prior to use, and checked against measurement standards traceable to international measurement standards where available, and proportionate to the risks identified.

- where components of the measuring systems cannot be calibrated, identify these in the Monitoring Plan and propose alternative control activities.
- when the equipment is found not to comply with required performance, the Project Proponent must promptly take necessary corrective action.

All measurements shall be conducted with calibrated measurement equipment according to industry standards including, where applicable:

- EN 14181 Stationary source emissions — Quality assurance of automated measuring systems;
- EN 15259 Air quality — Measurement of stationary source emissions — Requirements for measurement sections and sites and for the measurement objective, plan and report;
- ISO 16911-2 Stationary source emissions — Manual and automatic determination of velocity and volume flow rate in ducts.

Laboratories conducting analysis on behalf of the Project must be ISO 17025 accredited unless the Project Proponent can demonstrate this is not technically feasible or would incur unreasonable costs. The Project Proponent must demonstrate any non-accredited laboratory meets the requirements of ISO 17025 through ISO 9001 accreditation, other certified Quality Management System, or otherwise submit a PDD Appendix containing the following evidence for the laboratory:

- management of the personnel's competence for the specific tasks assigned;
- suitability of accommodation and environmental conditions;
- selection of analytical methods and relevant standards;

- where applicable, management of sampling and sample preparation, including control of sample integrity; where applicable, development and validation of new analytical methods or application of methods not covered by international or national standards;
- uncertainty estimation;
- management of equipment, including procedures for calibration, adjustment, maintenance and repair of equipment, and record keeping thereof;
- management and control of data, documents and software;
- management of calibration items and reference materials; quality assurance for calibration and test results, including regular participation in proficiency testing schemes, applying analytical methods to certified reference materials, or inter-comparison with an accredited laboratory;
- management of outsourced processes;
- management of assignments, customer complaints, and ensuring timely corrective action.

6.7.3
Climate Change Adaptation

Project Proponents must include in their PDD a robust climate risk and vulnerability assessment with the following steps:

- screening of the Project to identify which physical climate risks from the list in Section II of Appendix A of Regulation 2021/2139 may affect the performance of the Project during its expected lifetime;
- where the Project is assessed to be at risk from one or more of the physical climate risks listed in the same Appendix, a climate risk and vulnerability assessment to

assess the materiality of the physical climate risks on the Project;

- an assessment of adaptation solutions that can reduce the identified physical climate risk.

The climate risk and vulnerability assessment is proportionate to the scale of the Project and its expected lifespan, such that:

- for Projects with an expected lifespan of less than 10 years, the assessment is performed, at least by using climate projections at the smallest appropriate scale;
- for all other Projects, the assessment is performed using the highest available resolution, state-of-the-art climate projections across the existing range of future scenarios consistent with the expected lifetime of the activity, including, at least, 10 to 30 year climate projections scenarios for major investments;
- the range of future scenarios are the IPCC representative concentration pathways RCP2.6, RCP4.5, RCP6.0 and RCP8.5.

The climate projections and assessment of impacts are based on best practice and available guidance and take into account the state-of-the-art science for vulnerability and risk analysis and related methodologies in line with the most recent IPCC reports, scientific peer-reviewed publications, and open source or paid models.

For existing Projects and new Projects using existing physical assets, the Project Proponent must prepare and execute a plan for adaptation solutions, over a period of time of up to five years, that reduce the most important identified physical climate risks that are material to that Project.

For new Projects and existing Projects using newly-built physical assets, the Project Proponent must integrate the adaptation solutions that reduce the most important identified physical climate risks that are material to that activity at the time of design and construction before the start of operations.

The adaptation solutions implemented must not adversely affect the adaptation efforts or the level of resilience to physical climate risks of other people, of nature, of cultural heritage, of assets and of other economic activities; and be consistent with local, sectoral, regional or national adaptation strategies and plans; and consider the use of nature-based solutions or rely on blue or green infrastructure to the extent possible.

6.8
CRCF Glossary

CRCF - stands for Carbon Removals and Carbon Farming and is an EU-wide voluntary framework for certifying carbon removals, carbon farming and carbon storage in products.

CRCF Certificate of Compliance - A conformity statement issued by a certification body certifying that a project complies with Regulation (EU) 2024/3012; equivalent to an Isometric Verification Report in conjunction with the Project Validation Report.

CRCF Certified Unit - a Credit issued in accordance with Regulation (EU) 2024/3012

CRCF Activity - equivalent to an Isometric Project

CRCF Operator - equivalent to an Isometric Project Proponent

CRCF Certification Body - equivalent to a VVB

CRCF Certification Audit - equivalent to an Isometric Validation

CRCF Re-certification Audit - equivalent to an Isometric Verification

CRCF Activity Period - equivalent to an Isometric Crediting Period

CRCF Certification Period - equivalent to an Isometric Reporting Period

CRCF Activity Plan - equivalent to an Isometric Project Design Document (PDD)

CRCF Monitoring Plan - equivalent to the Monitoring Plan section of each Isometric Project Design Document (PDD)

CRCF Monitoring Report - equivalent to the GHG Statement submitted during Verification of Credits

7.0

Definitions and Acronyms

Approval Date
The date on which Isometric approves the initial Validation Report for a Project, following which the Project will be listed as Validated on the Registry. (Also known as "Registration Date").

Baseline
A set of data describing pre-intervention or control conditions to be used as a reference scenario for comparison.

Beneficiary
The organization benefiting from the Removal or Reduction claim afforded by a Credit. This may be the current holder of the Credit at the time of Retirement, or an organization specified by the Credit account holder during the Retirement procedure.

Cancellation (of a Credit)
The permanent annulling of a Credit to compensate for erroneous over-issuance or a Reversal. Once Canceled, the credit will no longer be available for Delivery or Retirement.

Carbon Flux
The amount of carbon exchanged between two or more Reservoirs over a period of time.

Certification (of a Protocol)
The Isometric process which involves expert review and Public Consultation in order to arrive at an approved version of a Protocol, against which Projects will be Validated and Removals or Reductions will be Verified.

Conservative
Purposefully erring on the side of caution under conditions of Uncertainty by choosing input parameter values that will result in a lower net CO_2 Removal or GHG Reduction than if using the median input values. This is done to increase the likelihood that a given Removal or

Reduction calculation is an underestimation rather than an overestimation.

Counterfactual
An assessment of what would have happened in the absence of a particular intervention – i.e., assuming the Baseline scenario.

Cradle-to-Grave
Considering impacts at each stage of a product's life cycle, from the time natural resources are extracted from the ground and processed through each subsequent stage of manufacturing, transportation, product use, and ultimately, disposal.

Credit
A publicly visible uniquely identifiable Credit Certificate Issued by a Registry that gives the owner of the Credit the right to account for one net metric tonne of Verified CO_2e Removal or Reduction. In the case of this Standard, the net tonne of CO_2e Removal or Reduction comes from a Project Validated against a Certified Protocol.

Credit Accounts
A function of the Isometric Registry that allows either Buyers or Project Proponents to receive Issued Credits, either by way of direct Credit Issuance in the case of Project Proponents, or by way of a Credit Transfer in the case of a Buyer.

Crediting Period
The period of time over which a Project Design Document is valid, and over which Removals or Reductions may be Verified, resulting in Issued Credits.

Delivery
The outcome of a Project Proponent providing Credits to fulfill Buyers' purchases.

Double Counting
Improperly allocating the same Removal or Reduction from a Project Proponent more than once to multiple Buyers.

Durability Threshold

A period of time, specified by a Protocol or Module, which Projects may select to use as a Project Durability Threshold. Where a Protocol or Module does not explicitly specify a Durability Threshold, the default Durability Threshold for the Protocol or Module is 1,000 years.

Environmental Additionality
An evaluation of the likelihood that an intervention causes a climate benefit above and beyond what would have happened in a no-intervention Baseline scenario.

Ex-ante Credits
Issuance of Credits prior to the carbon removal or GHG reduction taking place. Isometric does not Deliver Credits in this manner.

Ex-post Credits
Issuance of Credits after removal or reduction took place. This is the manner in which Isometric Delivers Credits.

GHG Statement
A document submitted alongside Claimed Removals and/or Reductions that details the calculations associated with a Removal or Reduction, including the Project's emissions, Removals, Reductions and Leakages, presented together in net metric tonnes of CO_2e per Removal or Reduction.

GHG Statement Report
Document submitted alongside a GHG Statement to provide supporting information and justification.

Greenhouse Gas (GHG)
Those gaseous constituents of the atmosphere, both natural and anthropogenic (human-caused), that absorb and emit radiation at specific wavelengths within the spectrum of terrestrial radiation emitted by the Earth's surface, by the atmosphere itself, and by clouds. This property causes the greenhouse effect, whereby heat is trapped in Earth's atmosphere (CDR Primer, 2022).

International Standards Organization (ISO)

A worldwide federation (NGO) of national standards bodies from more than 160 countries, one from each member country.

Isometric Crediting Program
The scope of Isometric which involves all of the processes relevant to the entire life cycle of a Credit.

Isometric Science Network
The Isometric group aimed at bringing together scientists working in industry, academia, NGOs and more to promote more efficient collaboration across the CDR ecosystem.

Isometric Systems
Isometric's proprietary code and processes that create/contain intellectual property (IP) and enable the Registry to function smoothly.

Issuance (of a Credit)
Credits are issued to the Credit Account of a Project Proponent with whom Isometric has a Validated Protocol after an Order for Verification and Credit Issuance services from a Buyer and once a Verified Removal or Reduction has taken place.

Leakage
The increase in GHG emissions outside the geographic or temporal boundary of a project that results from that project's activities.

Materiality
An acceptable difference between reported Removals/emissions or Reductions/emissions and what an auditor determines is the actual Removal/emissions or Reduction/emissions.

Model
A calculation, series of calculations or simulations that use input variables in order to generate values for variables of interest that are not directly measured.

Module
Independent components of Isometric Certified Protocols which are transferable between and applicable to different Protocols.

Monte Carlo Simulations
A mathematical approach for estimating the possible outcomes of an uncertain event through repeated random sampling. It can also be referred to as a "multiple probability simulation".

Non-Governmental Organization (NGO)
A nonprofit, usually with a societal, scientific, or political purpose; by definition an NGO is not associated with a governmental entity.

Offtake
A contract in which a Buyer agrees to purchase a set Removal and/or Reduction at a set price.

Project
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Project Design Document
The document, written by a Project Proponent, which records key characteristics of a Project and which forms the basis for Project Validation and evaluation in accordance with the relevant Certified Protocol. (Also known as “PDD”).

Project Design Document (PDD)
The document that clearly outlines how a Project will generate rigorously quantifiable Additional high-quality Removals or Reductions.

Project Durability Threshold
The period of time specified by a Project in its Project Design Document, which the Project must demonstrate Durability in excess of, and which the Durability of all Credits issued from a Project is equal to.

Project Proponent
The organization that develops and/or has overall legal ownership or control of a Removal or Reduction Project.

Project boundary
The defined temporal and geographical boundary of a Project.

Protocol
A document that describes how to quantitatively assess the net amount of CO_2 removed by a process. To Isometric, a Protocol is specific to a Project Proponent's process and comprised of Modules representing the Carbon Fluxes involved in the CDR process. A Protocol measures the full carbon impact of a process against the Baseline of it not occurring.

Proxy
A measurement which correlates with but is not a direct measurement of the variable of interest.

Public Consultation
The process by which a Protocol or Module is made available to the public for comment on the Isometric Registry.

Reduction
The term used to represent the reduction of greenhouse gasses emitted into the atmosphere from an existing emitter as a result of an emission reduction process.

Registry
A database that holds information on Verified Removals and Reductions based on Protocols. Registries Issue Credits, and track their ownership and Retirement.

Removal
The term used to represent the CO_2 taken out of the atmosphere as a result of a CDR process.

Reputable Source
A source that would be widely considered trustworthy based on the process undertaken (e.g., peer review) or origin of the information (e.g., government body).

Reservoir
A location where carbon is stored. This can be via physical barriers (such as geological formations) or through partitioning based on chemical or biological processes (such as mineralization or photosynthesis).

Retirement (of a Credit)
The act of confirming the final ownership of a given Credit and permanent withdrawal from circulation. The

Counterfactual of this act is the ultimate owner of the tonne of Removal or Reduction and the sole claimant as to its attributes for the purpose of carbon accounting.

Reversal
The escape of CO_2 to the atmosphere after it has been stored, and after a Credit has been Issued. A Reversal is classified as avoidable if a Project Proponent has influence or control over it and it likely could have been averted through application of reasonable risk mitigation measures. Any other Reversals will be classified as unavoidable.

Sensitivity Analysis
An analysis of how much different components in a Model contribute to the overall Uncertainty.

Sink
Any process, activity, or mechanism that removes a greenhouse gas, a precursor to a greenhouse gas, or an aerosol from the atmosphere.

Source
Any process or activity that releases a greenhouse gas, an aerosol, or a precursor of a greenhouse gas into the atmosphere.

Stakeholder
Any person or entity who can potentially affect or be affected by Isometric or an individual Project activity.

Standards (scientific)
Standard physical constants as well as standard values set forth by bodies such as the National Institute of Standards and Technology (NIST) or others.

System Boundary
GHG sources, sinks and reservoirs (SSRs) associated with the project boundary and included in the GHG Statement.

Uncertainty
A lack of knowledge of the exact amount of CO_2 removed by a particular process, Uncertainty may be quantified using probability distributions, confidence intervals, or variance estimates.

Validation

A systematic and independent process for evaluating the reasonableness of the assumptions, limitations and methods that support a Project and assessing whether the Project conforms to the criteria set forth in the Isometric Standard and the Protocol by which the Project is governed. Validation must be completed by an Isometric approved third-party (VVB).

Validation and Verification Bodies (VVBs)

Third-party auditing organizations that are experts in their sector and used to determine if a project conforms to the rules, regulations, and standards set out by a governing body. A VVB must be approved by Isometric prior to conducting validation and verification.

Verification

A process for evaluating and confirming the net Removals and Reductions for a Project, using data and information collected from the Project and assessing conformity with the criteria set forth in the Isometric Standard and the Protocol by which it is governed. Verification must be completed by an Isometric approved third-party (VVB).

8.0

Appendix A. Types of uncertainties

Calculation of Uncertainty should follow documented best practices, such as the requirements of the *Joint Committee for Guides in Metrology (JCGM) JCGM100:2008 - Evaluation of measurement data — Guide to the expression of uncertainty in measurement (GUM)*[19]. Uncertainty can be assessed using two different methods:

- **Type A** Uncertainty, where Uncertainty is assessed by making repeated observations of the parameter of interest
- **Type B** Uncertainty, where Uncertainty is founded on *a priori* distributions, evaluated by scientific judgment based on all of the available information on the possible variability of the parameter of interest

Type B Uncertainties may frequently be encountered, since it is not always practical to make repeated measurements for every input parameter, and many input parameters are published values or estimated through other means.

8.1

Emission factor uncertainty

Emission factors utilized should provide an Uncertainty value associated with the specific factor. Some databases from which emission factors are derived include and report the Uncertainty within the database[21]. The factor used, its source, and the value of the Uncertainty reported and used, should be documented. For emission factors for which no Uncertainty information is provided, clearly document this and refer to Unknown Uncertainties.

8.2

Measured parameter uncertainty

See the *GUM* for details on calculation of parameter or measurement Uncertainty. For instance, measurement parameter Uncertainty estimates could be derived from manufacturers specifications or calibration records. If repeated measurements are taken, the type A uncertainty should be reported as the standard deviation. Uncertainties should be documented and reported for each measured parameter.

8.3

Laboratory analysis uncertainty

For laboratory analyses, Uncertainty may be reported based on laboratory quality assurance documentation, including laboratory calibration results and records which use a reference calibration standard that is traceable to a national standard.

For a reported value based on multiple laboratory analyses, for which the value is reported as a mean, the type A uncertainty is calculated based on the standard deviation of the value around the mean.

8.4

Unknown uncertainties

When published Uncertainty information is not available and calculation via direct measurement is not feasible, estimation of Uncertainty should follow documented best practices, such as guidance from the *GUM* for evaluating type B Uncertainty. The specific approaches taken should be clearly documented.

Specifically regarding emission factors, the default values for basic Uncertainty from Section 10.1 of Ecoinvent's Overview and Methodology document may be used. Ecoinvent's protocol assumes a lognormal distribution for a value and then uses a Pedigree Matrix to assign uncertainties based on data quality.

8.5 Conservative estimate examples

Here are a couple of examples on how Option A (conservative estimate of input parameters) can be applied. These examples are for illustration purposes only:

Example 1: Uncertainty in Truck transportation carbon intensity

There are only a few carbon intensity factors published in a reputable emissions factor database for truck transportation. The factors range from 0.2 kg/km/t to 1.2 kg/km/t depending on different vehicle types, and the Project Proponent doesn't know exactly what type of vehicle is used. In this case, a higher carbon intensity leads to less net CO_2e Removal or Reduction and there are only a few relevant emissions factors to choose from so a distribution is not assumed. The project must take the conservative estimate to be the maximum value of 1.2 kg/km/t.

Example 2: Uncertainty in ocean DIC retention

For enhanced weathering in agriculture, a downstream loss term in the Removal calculation that cannot be directly measured is the percentage of bicarbonate ions from weathering that is retained as dissolved inorganic carbon (DIC) upon reaching the ocean. A Project Proponent uses a regional ocean model to determine that the DIC retention fraction follows a normal distribution, with a mean of 0.90

and a standard deviation of 0.02. A lower DIC retention fraction results in less net CO_2e Removal, so the conservative estimate is taken to be 1 standard deviation below the mean (corresponding to the 16th percentile). The conservative estimate of the DIC retention index is thus 0.88.

9.0

Appendix B: Stepwise process for uncertainty assessment

This appendix provides the detailed stepwise process for reducing and documenting uncertainty. The stepwise process must be completed alongside GHG Statements submitted for verification. Projects must demonstrate how they plan to reduce and document uncertainty as part of validation.

Step 1 - Input parameters information

Protocols must report a list of all parameters that are used in the Removal or Reduction calculations. Where uncertainty data is available from external sources, such as for emission factors or relating to equipment calibration, or where uncertainty information is determined from project measurements, this must be reported.

Step 2 - Contribution analysis

Separately, a Contribution Analysis should be undertaken that demonstrates the impact of each GHG SSR on the final net Removal or Reduction. This analysis should calculate the contribution of each GHG SSR by expressing the emissions associated with that SSR during the Reporting Period as a percentage of the total net removals for the same period.

Input parameters may be omitted from step 3 if the contribution analysis can demonstrate that the individual GHG SSR contributes to < 1% of net CO_2e Removal or Reduction and that all omitted GHG SSRs collectively contribute to < 1% of net CO_2e Removal or Reduction. For all other GHG SSRs, a Sensitivity Analysis is required.

Step 3 - Sensitivity analysis

A sensitivity analysis must be conducted to evaluate the impact of uncertainty in each input parameter on the final net CO_2e removal or reduction. The method used must be clearly documented so results can be independently reproduced.

Where one input parameter contributes to multiple GHG SSRs, this must be treated as one input parameter.

The purpose of the sensitivity analysis is to determine how uncertainty in an individual input parameter affects the net CO_2e removal or reduction estimate. This is done by adjusting the parameter using an appropriate uncertainty factor and recalculating the net CO_2e removal or reduction while keeping all other parameters constant.

If the resulting change in net CO_2e removal or reduction is less than 1%, no further action is required. If the change is equal to, or exceeds 1%, the adjusted value must be used in the GHG Statement. An exception to this rule is allowable for input parameters where data has been adopted to calculate GHG emissions and, the Project Proponent can demonstrate that:

- The input parameter is based on accurate, verifiable primary data, or
- The value used is conservative.

Any such justification must be clearly documented and supported by evidence.

When dealing with uncertainty, Project Proponents must follow the hierarchy outlined below to determine uncertainty factors:

1. Some Protocols contain specific guidance on assessing uncertainty in parameters. Project Proponents must first

consult the relevant protocol for any explicit rules on uncertainty estimation.

2. If no specific rules are included on uncertainty estimation in the relevant Protocol, then Project Proponents must follow the general guidance outlined below to assess uncertainty.
3. For input parameters based on primary data from direct measurements, the uncertainty factor must be derived using appropriate statistical methods to directly calculate the standard deviation of the parameter. These methods provide a systematic approach to quantifying uncertainty in the final net CO_2e removal or reduction estimate.

For parameters not based on primary data, the uncertainty factor may be informed by:

- Third-party data, such as meter accuracy from manufacturers or uncertainty ranges published with emission factors (e.g., in databases like Ecoinvent);
- Context-specific considerations, such as variability in logistics, seasonal effects, or operational practices;
- A default value of ±20%, only when no better information is available and with justification for its use.

All uncertainty factors must be clearly referenced, and the basis for their selection must be documented. Acceptable sources include instrument calibration data, published literature, expert judgment, proxies, and other reputable references. Additional guidance on assessing different types of uncertainties is provided in Appendix A.

The Sensitivity analysis must be repeated when any of the following conditions are met:

- The previous assessment was conducted one year or more prior to the current verification;
- There has been a significant change in the Net Removal or Reduction estimate;
- There has been a change in activities associated with any of the relevant SSRs.

10.0

Appendix C: Risk of Reversal Questionnaire

This questionnaire provides guidance on assessing Risk of Reversal, to be used by Protocols and Projects which need to make this assessment for use on the Isometric Registry.

This risk assessment identifies relevant material risk factors at the level of the Isometric Standard. The relevant risk factors from this Standard-level risk assessment are also specified at the Protocol level. Each relevant risk factor identified as part of a risk assessment must be included in the monitoring plan for a Project, and published as part of the Project Design Document.

The duration of required monitoring is determined in the relevant Protocol or Module. As above, the risk assessment informs the design of the monitoring plan requirements and consequently the extent, frequency and duration of monitoring.

Projects must re-assess their reversal risk using this questionnaire at the renewal of each crediting period, or sooner if:

- the relevant Protocol requires a more frequent assessment,
- monitoring identifies a reversal-related risk,
- the Project storage technology materially changes,
- an actual reversal event takes place.

If the relevant Protocol specifies that Reversals are not directly observable (i.e., the storage is an open system), the Project's Risk of Reversal is automatically "No observable

risk." Such Projects do not need to complete this questionnaire, but must still maintain a monitoring plan in accordance with the requirements of the relevant Protocol. Where a Protocol does not specify this, Projects must complete the Risk of Reversal questionnaire.
Please see the Reversals and Buffer Pools for more details on how Risk of Reversal is used.

	QUESTION	IF ANSWERED "YES"	IF ANSWERED "NO"
1	Is the carbon being stored in an impermeable geologic system? (e.g., salt cavern)	Proceed to questions 7-8	Add 1 to Risk Score and proceed to questions 2-9
2	Is the carbon being stored organic?	Add 1 to Risk Score	
3	Are conditions for methane production present (anaerobic conditions, lignin content)?	Add 1 to Risk Score	
4	Does this approach have a material risk of reversal due to natural disasters including, but not limited to, floods, storms, earthquakes, fires, etc.?	Add 1 to Risk Score	
5	Does this approach have a material risk of reversal due to human-induced events from outside actors, such as change in farming practices, change in ownership and management of project sites, or similar?	Add up to 2 to Risk Score	
6	**Applicable only for subsurface storage:** Is the carbon being stored with trapping mechanisms preventing reversals? (e.g., multiple confining layers, CO_2 dissolves or solidifies)	Minus 1 to Risk Score (unless 0)	
7	Is there 10+ years of monitoring and/or lab data demonstrating low project risk?	Minus up to 2 to Risk Score (unless 0)	
8	Does this pathway have a documented history of reversals in excess of proposed buffer pool size?	Add 2 to Risk Score	
9	Is there one or more project-specific factors that merit a high risk level?	Add up to 2 to Risk Score	

Risk Score Categories

Note the Risk Score at any step cannot be negative.

Where Reversals are not directly observable (as specified in the relevant Protocol), a Project's Risk of Reversal is automatically deemed "No observable risk", and no buffer contributions are required.

For other Projects, Risk of Reversal is determined based on the score of the risk assessment questionnaire, as follows:

- 0: Very Low Risk Level (1% buffer)
- 1-2: Low Risk Level (5% buffer)
- 3-4: Medium Risk Level (7% buffer)
- 5+: High Risk Level (10-20% buffer)

10.1 Worked examples

Isometric's Uncertainty and Buffer Pool policies represent two mostly non-overlapping approaches for how Isometric plans to be conservative in how credits are quantified and to potentially insure against future events that may result in a Reversal. Specifically, in removal pathways that utilize large, open-system reservoirs (e.g., the ocean) where direct observation of a reversal is not feasible, Isometric's Uncertainty policy will generally result in a larger uncertainty discount[22], and Isometric's Buffer Pool policy will generally result in a smaller Buffer Pool Size, or no Buffer Pool contribution. Conversely, in removal pathways that utilize closed-system reservoirs that are more straightforward to monitor, Isometric's Uncertainty policy will generally result in a smaller uncertainty discount, but the activity will be subject to a larger Buffer Pool Size. Below are two example scenarios that illustrate how and when these two policies are invoked.

Example 1: Ocean Alkalinity Enhancement

An OAE Project Proponent releases a known amount of alkalinity into the ocean. In accordance with the Protocol and PDD, the Project Proponent uses a model to estimate that the median CO_2 uptake after a certain amount of time is 100 tonnes. In accordance with Isometric's Uncertainty policy, the Project Proponent reports that the 16th percentile of CO_2 uptake outcomes is 15 tonnes below the median (resulting in a 15t uncertainty discount that is subtracted from the gross CO_2 removed). The Project Proponent does not need to complete a risk of reversal questionnaire, as Reversals are not directly observable for OAE, and as such the Project risk level is determined as “No observable risk” (resulting in no buffer pool contribution).

- Net CO_2e removed = 100t - 20t - 15t = 65t
  - Gross CO_2 removed: 100t
  - Counterfactual CO_2 removal and GHG emissions CO_2e: -20t
  - Uncertainty discount CO_2e: -15t
- Credits issued to Project Proponent: 63.7 credits (65 - 1.3)
  - Total Credits issued: 65 credits
  - Credits deposited to Buffer Pool account: 65 credits 0% = 0 credits

Example 2: Bio-oil Production and Injection

A bio-oil production and injection Project Proponent produces 100 tonnes of bio-oil with a well-constrained GHG Assessment and counterfactuals (there is no uncertainty discount for the amount of CO_2e removed because conservative parameters were used in the calculation). This bio-oil is then injected into an underground reservoir that is equipped with a suite of chemical sensors that can measure CO_2 and methane. The corresponding PDD has provisions to continue monitoring for fugitive gases for 40 years after

injection has concluded. The Project Proponent follows Isometric's guidance in answering the risk questionnaire and determines they have a Risk of Reversal that corresponds to Risk Level B (resulting in a 5% Buffer Pool Size).

- Net CO_2e removed = 100t - 20t - 0t = 80t
  - Gross CO_2 removed: 100t
  - Counterfactual CO_2 removal and GHG emissions CO_2e: -20t
- Credits issued to Project Proponent: 76 credits (80 - 4)
  - Total Credits Issued: 80 credits
  - Credits deposited to Buffer Pool account: 80 credits 5% = 4 credits

1. https://www.epa.gov/ghgemissions/overview-greenhouse-gases ↩
2. https://www.ipcc.ch/assessment-report/ar6/ ↩
3. https://www.iso.org/standard/66454.html ↩
4. https://www.ipcc.ch/report/ar6/wg3/downloads/report/IPCC_AR6_WGIII_Chapter12.pdf ↩ ↩[2]
5. https://rmi.org/insight/scaling-technological-greenhouse-gas-removal-a-global-roadmap-to-2050/ ↩
6. UK's Climate Change Committee (CCC) commissioned report Assessment of Options to Reduce Emissions from Fossil Fuel Production and Fugitive Emissions ↩
7. https://unfccc.int/process/parties-non-party-stakeholders/parties-convention-and-observer-states ↩
8. http://www.geol.lsu.edu/jlorenzo/geophysics/uncertainties/Uncertaintiespart2.html#muldiv ↩
9. Johansen, A.M., Evers, L. and Whiteley, N., 2010. Monte Carlo methods. International Encyclopedia of Education, pp.296-303. ↩
10. Section 4: Assessment Framework - Core Carbon Principles 2023, The Integrity Council for the Voluntary Carbon Market ↩ ↩[2]
11. International Labour Organisation Convention 169 on Indigenous and Tribal Peoples ↩
12. United Nations Declaration on the Rights of Indigenous Peoples, 13 September 2007 ↩
13. A negative impact is any adverse effect from a particular action that leaves an individual or community worse off, (i.e. where given a choice between the status quo and their position after that action, the individual would strictly prefer the status quo). ↩
14. UN Guiding Principles on Business and Human Rights Reporting Framework with implementation guidance (2015); International Business Leaders Forum (IBLF) and International Finance Corporation (IFC) Guide to Human

Rights Impact Assessment and Management (HRIAM), September 2011 ↩

15. FPIC must be obtained by following a suitable process. That process must identify the indigenous peoples concerned, document geographic and demographic mapping, involve a participatory communication approach which discloses project information transparently, reach consent, document any specific needs of indigenous peoples that must be included in the project, and include a feedback and complaints mechanism. ↩
16. International Bill of Human Rights, United Nations ↩
17. https://iaf.nu/en/accreditation-bodies/ ↩
18. ISO 14064-3:2019, 6.3.3 ↩
19. https://www.iso.org/sites/JCGM/GUM/JCGM100/C045315e-html/C045315e.html?csnumber=50461 ↩ ↩[2]
20. Regulation (EU) 2024/3012 ↩
21. Databases such as Ecoinvent provide uncertainty factors, often based on a lognormal probability distribution, with a 95% confidence interval. The reported uncertainty accounts for systematic error, random error, and may also use a pedigree matrix to assign additional qualitative data quality impacts as an uncertainty value. These uncertainties are reported as a combined standard uncertainty. See details and examples for ecoinvent: https://ecoinvent.org/wp-content/uploads/2020/10/dataqualityguideline*ecoinvent_3_20130506*.pdf ↩
22. "Uncertainty discount" is defined as the difference between the Conservative estimate of Removal, and the estimate of Removal in the absence of accounting for uncertainty ↩

Changelog

v2.0

CERTIFIED

Expanded scope to cover emissions reduction activities, including superpollutant projects (such as destruction of landfill methane emissions and HFCs). Introduced longer crediting periods; default crediting periods have been extended to 10 years, and to 15 years for DAC and Bio-CCS. Buffer pool requirements have been removed where these are not relevant for managing reversal risk: specifically for open systems (Biochar, Enhanced Rock Weathering, Ocean Alkalinity Enhancement), as well as any European projects where reversal liabilities are directly managed under the CCS Directive. Contributions have been reduced from 2% to 1% for "Very Low Risk" projects.

v1.9

CERTIFIED

Added CRCF section for alignment with the EU's Carbon Removal and Carbon Farming certification scheme. Introduced Non conformities and sanctions section. Introduced Prevention of scheme hopping section. Added definition of Vintage. Clarified shared Buffer Pools are defined in Biosphere Protocols.

v1.8

CERTIFIED

Clarify 'de facto' versus 'de jure' for regulatory additionality for UNFCCC Non-Annex I parties. Clarify life-cycle analysis calculation requirements. Introduce Appendix B: Stepwise process for uncertainty assessment. Update definitions of standard and protocol version numbers. Clarify protocol consultation requirements. Clarify protocol adoption requirements. Introduce protocol backward compatibility mechanisms. Introduce requirement for data collection and storage. Clarify No Double Counting also applies to claimed environmental attributes.

v1.7

CERTIFIED

Introduce Common Practice Analysis requirement. Clarifications to Financial Additionality requirements, Avoidable/Unavoidable Reversal requirements, PDD Documentation requirements and No Double Counting section.

v1.6

CERTIFIED

- Introduce language guide and corresponding wording clarifications
- Update CORSIA section regarding CORSIA-eligible emissions units
- Change handling of Durability, introducing definition of Protocol Durability Threshold(s) and Project Durability Threshold

v1.5

CERTIFIED

- Introduce a project public comment period.

- Clarify use of 'should' and 'must'.
- Add clarifying footnotes.

v1.4

CERTIFIED

- Update to the Buffer Pool Questionnaire.
- Size of Buffer Pool for Medium Risk Level is now 7% (previous 10%).

v1.3

CERTIFIED

- Update to maximum timing between verification events.
- Clarify expected reporting timeline for Projects to report Reversals.
- Remove all use of External (non-Isometric) Protocols.
- Include definition of Project "Approval Date" and clarify requirements regarding "Crediting Period Start Date".
- Include requirement for protecting against gender-based violence.
- Clarify requirement regarding provision of equal pay.
- Include reference to ILO-169 regarding IPLC rights.
- Update to Protocol versioning policy.
- Minor grammar and formatting updates.

v1.2

CERTIFIED

Initial release of v1.2

v1.1

CERTIFIED

Initial release of v1.1

v1.0

CERTIFIED

Initial release of v1.0

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